

**A PROJECT REPORT ON**  
**"QUICK COMMERCE OPERATIONAL INTELLIGENCE**  
**DASHBOARD"**

Submitted in the partial fulfillment of the requirement for the award of the Degree of

**BACHELOR OF COMPUTER APPLICATIONS**



**BENGALURU NORTH UNIVERSITY**

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## **CERTIFICATE**

This is to certify that the project entitled "QUICK COMMERCE OPERATIONAL INTELLIGENCE DASHBOARD" is a bonafide work done by M MOHAMMED FAHIM bearing Registration Number U19ON23S0137, in partial fulfillment for the award of Bachelor of Computer Applications of BENGALURU NORTH UNIVERSITY, during the academic year 2025-2026.

The report has not been submitted earlier either to this University / Institution for the fulfillment of the requirement of a course of study.

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## **STUDENT DECLARATION**

I, hereby declare that this report entitled "QUICK COMMERCE OPERATIONAL INTELLIGENCE DASHBOARD" is based on original work of independent research, carried out by me in partial fulfillment of Bachelor of Computer Applications Degree Course under BENGALURU NORTH UNIVERSITY of HKBK DEGREE COLLEGE under the guidance of PROF. SUBHAN SHAIK.

I also declare that this project is the outcome of my own efforts and that it has not been submitted to any other University or Institute for the award of any other Degree or Diploma or Certificate in BENGALURU NORTH UNIVERSITY or any other universities.

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## **ABSTRACT**

Quick commerce, also referred to as q-commerce or instant delivery, has emerged as one of the fastest growing segments in the modern retail industry. With consumer expectations now demanding delivery of essential goods within 10 to 30 minutes, companies such as Blinkit, Zepto, Swiggy Instamart, Dunzo, Amazon Now, Big Basket, Flipkart Minutes, and Jio Mart are competing intensely to capture urban markets across India, particularly in metropolitan cities like Bengaluru.

This project presents the Quick Commerce Operational Intelligence Dashboard, a comprehensive data analytics and machine learning-powered web application built using Python, Flask, Pandas, NumPy, Scikit-learn, HTML, CSS, and Chart.js/Plotly. The dashboard is designed to analyze, visualize, and predict key operational parameters that directly impact delivery efficiency, customer satisfaction, revenue generation, and infrastructure utilization across the quick commerce ecosystem.

The dataset used in this project comprises 45,301 real-world quick-commerce order records spanning multiple companies and urban regions across Bengaluru. The data encompasses attributes such as delivery time, order value, customer rating, number of dark stores, demand per region, peak hour indicators, distance, number of items, and company-specific information.

The core analytics modules include: (1) Delivery Time Prediction using Linear Regression, which forecasts expected delivery time based on distance, order hour, number of items, company, and city; (2) Discount Prediction using Logistic Regression, which classifies whether an order is likely to receive a discount based on multiple operational features; (3) Peak Hour Analysis, which identifies high-demand operational windows; (4) Dark Store Infrastructure Analysis, which evaluates the impact of dark store density on delivery efficiency; (5) Revenue and Order Volume Analysis by company and region; and (6) Customer Rating Analysis across companies and delivery performance tiers.

Key findings of the dashboard include: Zepto demonstrates the fastest average delivery time despite maintaining relatively high operational demand, indicating efficient infrastructure

utilization. Blinkit records the highest customer satisfaction ratings. Rajajinagar emerges as the most operationally overloaded region in Bengaluru. Peak delivery demand is observed during the 12:00–14:00 and 19:00–21:00 hours. A clear inverse relationship exists between dark store density and average delivery time, confirming the strategic importance of micro-fulfillment center distribution.

The dashboard also incorporates a Business Recommendations module that provides company-specific and region-specific actionable intelligence, along with a comprehensive Company Insights section. The web application is built on a Flask-based backend with interactive front-end visualizations, supporting real-time prediction and dynamic data exploration through an intuitive user interface.

This project demonstrates how the integration of data science, business analytics, and predictive modeling can support strategic decision-making in the modern quick commerce ecosystem, enabling operational optimization and competitive differentiation.

**Keywords:**

Quick Commerce, Operational Intelligence, Dark Store Analytics, Delivery Time Prediction, Flask Dashboard, Logistic Regression, Linear Regression, Business Intelligence, Demand Forecasting, Bengaluru.

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# **CHAPTER 1**



# 1: INTRODUCTION

## 1.1 Background and Motivation

The retail industry has undergone a dramatic transformation over the past decade, with e-commerce evolving from multi-day delivery models to same-day and, more recently, ultra-fast delivery paradigms. Quick commerce, commonly abbreviated as q-commerce, represents the latest frontier in this evolution, characterized by the promise of delivering everyday essential items—groceries, personal care products, household supplies, and beverages—within a compressed time window of 10 to 30 minutes from the moment an order is placed.

India, with its rapidly growing urban middle class, widespread smartphone adoption, and dense urban neighborhoods, has emerged as one of the most competitive quick commerce markets globally. Companies such as Blinkit (formerly Grofers), Zepto, Swiggy Instamart, Dunzo, Amazon Now, Big Basket, Flipkart Minutes, and Jio Mart are actively competing for market share across Indian metropolitan cities, particularly Bengaluru, which serves as a technology and innovation hub with a young, tech-savvy, and high-income consumer base.

The operational backbone of quick commerce is the concept of the dark store—a micro-fulfillment center, also referred to as a warehouse or dark store, strategically located within 2 to 5 kilometers of dense residential neighborhoods. Unlike traditional retail stores that serve walk-in customers, dark stores are exclusively dedicated to fulfilling online orders through a team of personal shoppers and a fleet of delivery riders. The density, location, and operational efficiency of these dark stores directly determine the speed and reliability of deliveries.

Despite the explosive growth of the quick commerce sector, companies face significant operational challenges. These include managing uneven demand across different urban regions, identifying overloaded dark stores that strain capacity during peak hours, optimizing delivery routes and personnel allocation, and maintaining high customer satisfaction ratings while scaling operations. The absence of a centralized, data-driven operational intelligence platform makes it difficult for operations managers and business analysts to make informed decisions in real time.

This project was motivated by the recognition that the vast volumes of operational data generated by quick commerce platforms—order records, delivery times, customer ratings, demand patterns, and regional statistics—contain invaluable insights that, if properly analyzed and visualized, could significantly enhance operational efficiency, reduce costs, and improve customer experience.

The rapid rise of digital commerce and changing customer expectations have transformed the traditional retail and delivery ecosystem into a highly time-sensitive operational environment. Consumers today expect groceries, medicines, personal care products, beverages, and household essentials to be delivered within a few minutes of placing an order. This shift has created immense operational pressure on quick commerce companies to optimize logistics, infrastructure, and delivery operations.

Unlike traditional e-commerce platforms that prioritize inventory variety and scheduled deliveries, quick commerce platforms focus primarily on speed, operational efficiency, and hyperlocal fulfillment strategies. This operational model introduces multiple challenges such as dark store placement optimization, delivery route efficiency, rider allocation, peak-hour traffic management, and customer satisfaction maintenance.

As operational complexity increases, quick commerce companies generate large volumes of transactional and operational data on a daily basis. However, without a centralized analytical system, extracting meaningful insights from this data becomes difficult. Business teams often struggle to identify overloaded delivery regions, inefficient infrastructure allocation, and operational bottlenecks that negatively affect customer experience.

The Quick Commerce Operational Intelligence Dashboard addresses these challenges by integrating data analytics, machine learning, and business intelligence into a single unified platform. The system not only visualizes operational performance but also provides predictive insights that support proactive decision-making and infrastructure planning.

## 1.2 Problem Statement

Quick commerce platforms generate enormous volumes of transactional and operational data on a daily basis. However, this data remains largely underutilized due to the absence of integrated analytical tools capable of processing, visualizing, and deriving actionable intelligence from it in a structured and user-friendly manner.

The key operational problems identified in the quick commerce domain include:

- **Regional demand imbalance:** Certain urban zones, such as Rajajinagar, Whitefield, and Banashankari, experience disproportionately high order volumes relative to the number of dark stores present in those areas, resulting in capacity pressure and increased delivery times.
- **Inefficient peak-hour resource allocation:** The lack of predictive models for delivery time during peak operational hours (12:00 PM–2:00 PM and 7:00 PM–9:00 PM) leads to suboptimal staffing and vehicle deployment.
- **Limited infrastructure visibility:** Companies lack consolidated dashboards to monitor dark store utilization, compare performance across regions, and identify infrastructure gaps at a glance.
- **Customer satisfaction variability:** Customer ratings vary significantly across companies and regions, and the root causes of rating differentials are not systematically analyzed or addressed.
- **Revenue optimization gaps:** Revenue and order volume patterns across companies and cities are not monitored in an integrated manner, limiting strategic business planning.

This project addresses these problems by developing an integrated Operational Intelligence Dashboard that consolidates data analytics, machine learning predictions, and business intelligence into a single, interactive web-based platform.

## 1.3 Objectives of the Project

The primary objective of this project is to develop an interactive web-based Operational Intelligence Dashboard for the quick commerce sector using Python, Flask, Pandas, NumPy, Scikit-learn, and various data visualization libraries. The project aims to perform comprehensive Exploratory Data Analysis (EDA) on a dataset containing 45,301 quick commerce orders to identify operational patterns, trends, anomalies, and business opportunities. Another key objective is to develop a Linear Regression model capable of predicting delivery time based on factors such as distance, order hour, number of items, company, and geographic region. In addition, a Logistic Regression model is implemented to predict whether an order is likely to qualify for a discount, thereby supporting promotional planning and business decision-making.

The project further focuses on analyzing peak-hour demand patterns to identify critical time windows that require additional operational resources and workforce allocation. It also evaluates

the relationship between dark store density and delivery efficiency, enabling businesses to optimize infrastructure placement and reduce delivery delays. Through advanced analytics and visualization techniques, the system examines revenue distribution, order volume trends, and customer rating variations across multiple companies and regions. Based on these insights, the dashboard generates company-specific and region-specific business recommendations to support strategic decision-making. Furthermore, the project provides an intuitive and interactive user interface that enables operations managers, business analysts, and stakeholders to explore operational data, visualize key performance indicators, and generate real-time predictions efficiently.

## 1.4 Scope of the Project

The scope of this project covers multiple aspects of operational analytics and machine learning within the quick commerce industry. Geographically, the analysis focuses on several major urban regions of Bengaluru, including Whitefield, Indiranagar, HSR Layout, Koramangala, BTM Layout, Marathahalli, Electronic City, Banashankari, and Rajajinagar, among others. The study incorporates operational data from eight leading quick commerce companies, namely Blinkit, Zepto, Swiggy Instamart, Dunzo, Amazon Now, Big Basket, Flipkart Minutes, and Jio Mart, enabling a comprehensive comparative analysis of delivery performance, customer satisfaction, and business operations.

The project encompasses a wide range of analytical modules, including delivery time analysis, dark store utilization assessment, peak-hour demand analysis, revenue and order volume evaluation, customer rating benchmarking, machine learning-based predictions, and business recommendation generation. The system is developed using a modern technology stack comprising Python 3.x, Flask, Pandas, NumPy, Scikit-learn, HTML5, CSS3, JavaScript, Chart.js, and Plotly to ensure efficient data processing, predictive modeling, and interactive visualization.

The analysis is performed on a dataset consisting of 45,301 quick commerce order records, providing a substantial foundation for extracting meaningful business insights. Furthermore, the project includes predictive models for delivery time estimation and discount eligibility classification using trained machine learning algorithms. Although the current implementation does not integrate with real-time APIs or live company systems, the architecture supports future enhancements through dataset updates and model retraining to improve predictive accuracy and operational relevance.

## 1.5 Organization of the Report

This report is organized into eleven chapters as follows:

- Chapter 1 – Introduction: Provides the background, motivation, problem statement, objectives, and scope of the project.
- Chapter 2 – Literature Survey: Reviews existing research and commercial systems in the domain of delivery optimization, dark store analytics, and operational dashboards.
- Chapter 3 – Existing and Proposed System: Compares the limitations of current approaches with the proposed solution.

- Chapter 4 – Feasibility Study: Assesses the technical, economic, and operational feasibility of the project.
- Chapter 5 – System Requirements: Details the hardware and software requirements for development and deployment.
- Chapter 6 – System Design: Presents the system architecture, UML diagrams, ER diagram, DFD, context diagram, sequence diagram, and database design.
- Chapter 7 – Methodology: Describes the development approach, data preprocessing, feature engineering, ML workflow, Flask application architecture, and dashboard development.
- Chapter 8 – Implementation: Details the module-wise implementation with code excerpts and module descriptions.
- Chapter 9 – Results and Discussion: Presents the dashboard outputs, model performance metrics, key analytical findings, and business recommendations.
- Chapter 10 – Advantages, Limitations, and Future Enhancements: Discusses the strengths, current limitations, and future roadmap of the system.
- Chapter 11 – Conclusion: Summarizes the project achievements and contributions.

# **CHAPTER 2**

## **2: LITERATURE SURVEY**

### **2.1 Review of Existing Research and Systems**

#### **2.1.1 Delivery Time Prediction and Route Optimization**

The application of machine learning techniques to predict delivery time and optimize logistics routes has been an active area of research over the past decade. Ruan et al. (2020) [1] proposed an ensemble learning framework that combined gradient boosting with neural networks for last-mile delivery time estimation, achieving a mean absolute error (MAE) of 2.3 minutes on urban delivery datasets. The study highlighted the significant impact of distance, traffic density, and time-of-day features on prediction accuracy.

Voccia, Campbell, and Thomas (2019) [2] explored same-day delivery models and dynamic vehicle routing algorithms, demonstrating that micro-fulfillment center placement within 3 kilometers of customer clusters could reduce average delivery time by up to 35%. Their simulation-based study provided a foundational framework for dark store location optimization that informs the infrastructure analysis components of this project.

Wang and Chen (2021) [3] developed a real-time delivery time prediction system using Long Short-Term Memory (LSTM) neural networks, incorporating live traffic data and order characteristics. While their approach achieved superior performance, the computational complexity and real-time data dependency make it less suitable for the offline analytical context of this project. The current project employs Linear Regression as a more interpretable and computationally efficient alternative, appropriate for the academic context and offline dataset used.

#### **2.1.2 Dark Store and Micro-Fulfillment Center Analytics**

The strategic placement and operational efficiency of dark stores (micro-fulfillment centers) has been studied extensively in the context of omnichannel retail. Galino and Moreno (2022) [4] demonstrated through field experiments across European grocery retailers that adding a dark store within a 2 km radius of high-demand zones reduced average order fulfillment time by 28% and improved customer retention by 12%.

Song et al. (2021) [5] proposed a mixed-integer programming model for dark store network optimization, balancing fixed infrastructure costs against the variable operational costs of longer delivery distances. Their findings confirm the intuition validated in this project: that regions with fewer dark stores relative to demand experience significantly higher delivery times and lower customer satisfaction.

In the Indian context, Mehta and Gupta (2023) [6] analyzed the dark store expansion strategies of Blinkit and Zepto in Bengaluru, finding that Zepto's strategy of clustering multiple small dark stores in high-demand neighborhoods outperformed Blinkit's larger, more dispersed store model in terms of delivery speed during peak hours. These findings align with the key insight from this project's dashboard that Zepto achieves the lowest average delivery time despite high operational demand.

### **2.1.3 Business Intelligence Dashboards for Logistics**

Business intelligence (BI) dashboards have become essential tools for operational decision-making in logistics and supply chain management. Isik, Jones, and Sidorova (2013) [7] established a theoretical framework for BI system success, identifying data quality, system quality, and information quality as the three primary determinants of decision-making effectiveness. Their framework guided the design principles of the dashboard developed in this project.

Power (2016) [8] reviewed the evolution of decision support systems in logistics, noting the transition from static report-generation tools to dynamic, interactive dashboards that support real-time exploration of operational data. The current project implements an interactive web-based dashboard aligned with this modern paradigm, using Flask as the application server and Chart.js/Plotly for dynamic visualizations.

Sharma and Verma (2022) [9] developed a logistics analytics dashboard for Indian e-commerce platforms using Python and Dash, covering delivery performance, return rate analysis, and zone-wise demand visualization. While their work shares conceptual similarities with this project, it does not incorporate predictive machine learning models or dark store infrastructure analysis, which are key differentiating features of this project.



### **2.1.4 Customer Rating and Satisfaction Analytics**

Customer satisfaction in the quick commerce domain is closely tied to delivery speed, order accuracy, and product quality. Kannan and Li (2017) [10] demonstrated through regression analysis that a one-minute reduction in delivery time corresponds to a 0.08-point improvement in customer ratings on a 5-point scale, a finding consistent with the positive correlation between delivery efficiency and ratings observed in this project's dashboard.

Zhang et al. (2020) [11] applied sentiment analysis and machine learning to customer reviews from food delivery platforms, identifying delivery time, packaging quality, and customer service responsiveness as the three most significant determinants of customer satisfaction. The customer rating analysis module in this project quantifies satisfaction differentials across companies, providing a foundation for targeted service improvement recommendations.

### **2.1.5 Peak Hour Demand Forecasting**

Understanding and predicting peak demand periods is critical for resource allocation in time-sensitive delivery services. Taylor (2012) [12] proposed a seasonal exponential smoothing model for intraday demand forecasting in service operations, demonstrating that operational planning aligned with predicted demand peaks reduces service time variability by up to 22%. The peak hour analysis component of this project adopts a descriptive statistical approach to identify demand peaks, which can serve as a foundation for future predictive extensions.

Recent advancements in business intelligence systems and predictive analytics have significantly influenced logistics and operational management across various industries. Companies increasingly rely on data-driven systems to improve operational visibility, reduce delays, optimize infrastructure usage, and improve customer satisfaction.

Machine learning algorithms such as Linear Regression, Logistic Regression, Random Forest, Gradient Boosting, and Neural Networks are commonly used in logistics prediction systems. These models help organizations estimate delivery times, forecast demand, predict operational overload, and automate decision-making processes.

Modern dashboard systems have evolved from static reporting tools into dynamic analytical platforms capable of supporting real-time business decisions. Interactive dashboards allow

organizations to monitor KPIs, identify anomalies, and visualize operational patterns through graphs, charts, and predictive insights.

Despite these technological advancements, limited research exists on integrated operational intelligence systems specifically designed for the Indian quick commerce ecosystem. Most existing systems focus only on delivery optimization or demand forecasting individually, without combining infrastructure analysis, customer satisfaction analytics, revenue analysis, and predictive modeling into a single unified platform.

This project attempts to bridge this gap by developing a comprehensive dashboard tailored specifically for Indian quick commerce operations using open-source technologies and machine learning techniques.

## **2.2 Research Gap Identification**

The literature review conducted for this project highlights several research gaps within the quick commerce and logistics analytics domain. Existing studies primarily focus on individual analytical problems such as delivery time prediction, dark store optimization, customer satisfaction analysis, or demand forecasting in isolation. However, there is limited research on integrated systems that combine these analytical dimensions into a single operational intelligence platform. This project addresses that gap by developing a unified dashboard that brings together analytics, visualization, predictive modeling, and business recommendations specifically for the quick commerce sector.

Another significant gap identified in the literature is the lack of multi-company comparative analysis. Most existing solutions are designed for a single organization or are developed for generic logistics applications without providing direct performance comparisons among competing companies. This project overcomes this limitation by analyzing and comparing operational data from eight major quick commerce companies within a common analytical framework, enabling meaningful benchmarking and performance evaluation.

Furthermore, many existing analytics systems are either proprietary enterprise solutions with limited accessibility or academic prototypes that lack practical usability and interactive interfaces. To address this challenge, the proposed system provides a fully functional, user-friendly web dashboard developed using open-source technologies, allowing users to explore insights, generate predictions, and visualize operational performance through an intuitive interface. Additionally, the majority of existing research is based on global logistics datasets and may not accurately reflect the operational characteristics of the Indian market. This project specifically focuses on Bengaluru and the Indian quick commerce ecosystem, making the findings, recommendations, and predictive models highly relevant to real-world urban delivery operations in India.

# **CHAPTER 3**

## **3: EXISTING SYSTEM AND PROPOSED SYSTEM**

### **3.1 Existing System**

Currently, quick commerce companies rely on a combination of siloed internal tools, standalone spreadsheets, and limited business intelligence platforms to monitor their operational performance. The following systems represent the existing approaches in the domain:

#### **3.1.1 Manual Spreadsheet Reporting**

Many quick commerce operations teams use Microsoft Excel or Google Sheets to aggregate daily delivery metrics, compute averages, and generate periodic reports. While these tools provide basic analytical capability, they are manual, error-prone, not scalable to tens of thousands of records, and do not support real-time visualization or predictive modeling.

#### **3.1.2 Generic BI Tools**

Enterprise BI tools such as Tableau, Power BI, and Looker are used by some larger quick commerce companies for dashboard visualization. However, these tools require expensive licensing, specialized BI expertise for configuration, and do not natively support custom machine learning model integration. Additionally, they are not tailored to the specific operational dynamics of quick commerce.

#### **3.1.3 Platform-Specific Dashboards**

Individual quick commerce platforms maintain internal operational dashboards, but these are proprietary, inaccessible for cross-platform comparison, and not designed for the integrated analytical perspective required by operations researchers, business analysts, or academic study.

#### **3.1.4 Academic Prototype Systems**

Academic research has produced prototype systems for delivery optimization and demand forecasting, but these are typically not implemented as deployable web applications, lack interactive interfaces, and are not integrated with real-world multi-company datasets.

### **3.2 Disadvantages of Existing System**

The analysis of existing systems revealed several limitations that motivated the development of the proposed Quick Commerce Operational Intelligence Dashboard. One of the major challenges is the lack of integration among analytical tools. In many organizations, operational insights are distributed across multiple software platforms, spreadsheets, and reporting systems, making it difficult for decision-makers to obtain a comprehensive view of business performance. This fragmented approach often leads to inefficiencies in monitoring delivery operations, customer satisfaction, and infrastructure utilization.

Another significant limitation is the absence of predictive capabilities within traditional reporting and business intelligence tools. While existing spreadsheet-based and dashboard solutions provide descriptive analytics and historical reporting, they generally lack built-in machine learning functionality for forecasting delivery times or predicting discount eligibility. As a result, organizations are often unable to anticipate operational challenges proactively and must rely on retrospective analysis.

The high cost and complexity associated with enterprise-level business intelligence platforms present another barrier. Many commercial analytics solutions require expensive licensing fees, specialized technical expertise, and dedicated infrastructure, making them inaccessible to smaller operations teams, startups, and academic researchers. Furthermore, most existing systems are designed to operate within a single organization and do not provide mechanisms for cross-company benchmarking. This limitation prevents stakeholders from comparing operational performance across competing quick commerce platforms and identifying industry best practices.

Additionally, existing analytics tools offer limited support for dark store intelligence and infrastructure optimization. Although dark stores play a critical role in enabling rapid order fulfillment, few accessible solutions provide integrated analysis of dark store utilization and its impact on delivery performance.

Another common issue is the limited actionability of available analytics platforms. Many systems present large volumes of data through reports and visualizations but fail to generate structured, company-specific recommendations that can support strategic decision-making.

Finally, scalability remains a challenge for traditional manual reporting methods, which often struggle to efficiently process datasets containing tens of thousands of records and deliver timely insights. The proposed system addresses these limitations by integrating analytics, machine learning, visualization, and recommendation generation into a single scalable web-based platform capable of handling large operational datasets and supporting data-driven decision-making.

### **3.3 Proposed System**

The proposed Quick Commerce Operational Intelligence Dashboard addresses all the limitations of the existing systems by providing a unified, open-source, web-based platform with the following key capabilities:

The proposed Quick Commerce Operational Intelligence Dashboard offers a comprehensive and integrated solution for analyzing operational performance within the quick commerce ecosystem. One of its key strengths is the integration of multiple analytical modules into a single Flask-based web application. The system consolidates delivery analysis, dark store analytics, peak-hour demand analysis, revenue visualization, customer rating benchmarking, machine learning predictions, and business recommendation generation into a unified platform. This integrated approach eliminates the need for multiple disconnected tools and enables users to access all critical insights from a centralized dashboard.

A significant feature of the proposed system is the incorporation of machine learning models directly within the web interface. The dashboard utilizes a Linear Regression model for delivery time prediction and a Logistic Regression model for discount prediction, allowing users to generate forecasts on demand using operational parameters. These predictive capabilities enhance decision-making by enabling organizations to anticipate delivery performance and promotional outcomes before they occur.

The system also provides highly interactive visualizations through Chart.js and Plotly, enabling users to explore data dynamically through features such as hovering, zooming, and filtering. These visualizations improve data interpretation and make complex operational patterns easier to understand. In addition, the dashboard supports multi-company comparative analytics by allowing side-by-side performance evaluation of eight major quick commerce companies using a unified dataset. This functionality facilitates benchmarking and helps identify operational strengths and weaknesses across competitors.

Another distinguishing feature is the inclusion of dark store infrastructure intelligence. The system analyzes and visualizes the relationship between dark store density and delivery performance, enabling organizations to make informed infrastructure planning decisions. Furthermore, the business recommendations module generates structured, data-driven suggestions tailored to specific companies and regions, thereby supporting strategic planning and operational optimization.

The proposed system is built entirely using open-source technologies, making it cost-effective, easily deployable, and highly customizable without licensing restrictions. Its scalable architecture, based on Flask and Pandas, supports efficient processing of large datasets and provides a strong foundation for future enhancements, including real-time data integration, cloud deployment, and advanced predictive analytics.

### 3.4 Advantages of Proposed System

| Feature                  | Existing System         | Proposed System              |
|--------------------------|-------------------------|------------------------------|
| Data Integration         | Siloed, manual          | Unified, automated           |
| Predictive Models        | None                    | Linear & Logistic Regression |
| Visualization            | Static charts           | Interactive, dynamic         |
| Cost                     | High (enterprise tools) | Zero (open-source)           |
| Cross-Company Analysis   | Not supported           | Fully supported              |
| Dark Store Analytics     | Absent                  | Integrated                   |
| Business Recommendations | Manual interpretation   | Automated, structured        |
| Scalability              | Low                     | High                         |
| User Interface           | Generic BI              | Custom, domain-specific      |
| Deployment               | Complex                 | Flask, lightweight           |

# **CHAPTER 4**

## **4: FEASIBILITY STUDY**

A feasibility study is conducted to assess whether the proposed system can be implemented successfully within the given constraints of time, resources, and technology. The three principal dimensions of feasibility are technical, economic, and operational.

### **4.1 Technical Feasibility**

Technical feasibility evaluates whether the required technologies, tools, and infrastructure are available and capable of supporting the proposed system.

#### **4.1.1 Technology Availability**

All technologies required for this project are freely available, mature, and well-documented. Python 3.x provides a robust programming environment with extensive libraries for data science and machine learning. Flask is a lightweight, widely-used web framework that supports rapid application development. Pandas and NumPy provide comprehensive data manipulation capabilities. Scikit-learn offers a complete suite of machine learning algorithms including Linear and Logistic Regression. Chart.js and Plotly provide mature, browser-compatible visualization libraries.

#### **4.1.2 Dataset Availability**

The dataset of 45,301 quick commerce order records has been compiled and preprocessed for this project. The dataset includes all required attributes for the planned analyses and ML models, including order time, delivery distance, company, city, customer rating, order value, dark store count, and demand metrics.

#### **4.1.3 Development Infrastructure**

The project can be developed on a standard laptop with 8 GB RAM, a dual-core processor, and internet connectivity for library installations. No specialized hardware or cloud infrastructure is required for the development and demonstration phase.

#### **4.1.4 Technical Conclusion**

The project is technically feasible. All required technologies are available, proven, and within the technical competency of the development team. The dataset is of sufficient size and quality to support the planned analytics and ML models.



## **4.2 Economic Feasibility**

Economic feasibility evaluates whether the cost of developing and maintaining the proposed system is justified by the benefits it delivers.

### **4.2.1 Development Cost**

All software, libraries, and frameworks used in this project are open-source and freely available, resulting in zero software licensing costs. The development hardware (personal laptop) is already available, incurring no additional capital expenditure. The primary cost is the developer's time, estimated at approximately 180–220 person-hours across all phases of the project.

### **4.2.2 Operational Benefits**

For actual deployment in a quick commerce company, the dashboard could provide significant cost savings through: (a) improved resource allocation during peak hours, estimated to reduce delivery personnel idle time by 15–20%; (b) dark store placement optimization, reducing infrastructure costs; (c) customer satisfaction improvement, reducing churn and support costs.

### **4.2.3 Economic Conclusion**

The project is economically feasible for both academic purposes (near-zero cost) and potential commercial deployment (high return on investment relative to development cost).

## **4.3 Operational Feasibility**

Operational feasibility evaluates whether the system will be used effectively once it is deployed, considering user acceptance, organizational impact, and ease of use.

### **4.3.1 User Interface Accessibility**

The web-based interface is designed to be intuitive and accessible to non-technical users, including operations managers and business analysts, through a standard web browser. No specialized training or technical knowledge is required to navigate the dashboard or generate predictions.

### **4.3.2 Organizational Fit**

The system is designed to complement existing operational workflows rather than replace them. It provides analytical intelligence that supports, rather than substitutes, human decision-making.

### **4.3.3 Operational Conclusion**

The project is operationally feasible. The web-based interface, intuitive navigation, and alignment with existing decision-making workflows ensure high likelihood of adoption by intended users.

# **CHAPTER 5**

## 5: SYSTEM REQUIREMENTS

### 5.1 Hardware Requirements

| Component        | Minimum Requirement         | Recommended Specification         |
|------------------|-----------------------------|-----------------------------------|
| Processor        | Intel Core i3 / AMD Ryzen 3 | Intel Core i5/i7 or AMD Ryzen 5/7 |
| RAM              | 4 GB                        | 8 GB or more                      |
| Hard Disk        | 20 GB free space            | 50 GB SSD                         |
| Display          | 1024 × 768 resolution       | 1920 × 1080 Full HD               |
| Network          | Internet access for setup   | Broadband (10 Mbps+)              |
| Operating System | Windows 10 / Ubuntu 18.04   | Windows 11 / Ubuntu 22.04         |

### 5.2 Software Requirements

| Software / Library | Version                    | Purpose                             |
|--------------------|----------------------------|-------------------------------------|
| Python             | 3.9 or above               | Core programming language           |
| Flask              | 2.3.x                      | Web application framework           |
| Pandas             | 1.5.x / 2.0.x              | Data manipulation and analysis      |
| NumPy              | 1.24.x                     | Numerical computations              |
| Scikit-learn       | 1.2.x                      | Machine learning models             |
| Matplotlib         | 3.7.x                      | Data visualization (backend)        |
| Seaborn            | 0.12.x                     | Statistical visualization           |
| Chart.js           | 4.x                        | Interactive frontend charts         |
| Plotly             | 5.x                        | Advanced interactive visualizations |
| HTML5 / CSS3       | Latest                     | Frontend structure and styling      |
| JavaScript (ES6+)  | Latest                     | Frontend interactivity              |
| Jinja2             | 3.1.x                      | Flask templating engine             |
| pip                | 23.x                       | Python package management           |
| Visual Studio Code | 1.80+                      | Integrated Development Environment  |
| Git                | 2.40+                      | Version control                     |
| Web Browser        | Chrome 115+ / Firefox 115+ | Dashboard access                    |

## **5.3 Functional Requirements**

### **5.3.1 Data Management**

- The system shall load and process the quick commerce order dataset (CSV format) on application startup.
- The system shall perform automated data cleaning and preprocessing, including handling of missing values and data type conversion.
- The system shall support reloading of updated datasets without requiring code modification.

### **5.3.2 Analytics and Visualization**

- The system shall display a summary dashboard with key performance metrics (total orders, best-rated company, most overloaded region, fastest delivery company).
- The system shall provide interactive charts for delivery analysis, dark store performance, revenue analysis, peak hour patterns, and customer rating benchmarking.
- The system shall display the top 10 overloaded dark store regions, operational load vs. delivery efficiency scatter chart, and order value vs. customer rating correlation.

### **5.3.3 Prediction Module**

- The system shall accept user inputs for delivery time prediction (distance, number of items, order hour, company, city) and return a predicted delivery time.
- The system shall accept user inputs for discount prediction (number of items, order hour, city, company, order day) and return a discount eligibility prediction.

### **5.3.4 Business Intelligence**

- The system shall display company-specific business recommendations based on analytical findings.
- The system shall present company insights categorized by customer rating, dark store infrastructure, operational efficiency, peak hour, and revenue trends.

## **5.4 Non-Functional Requirements**

The proposed Quick Commerce Operational Intelligence Dashboard is designed to satisfy several important non-functional requirements that ensure efficient system performance, usability, reliability, maintainability, scalability, portability, and security. From a performance perspective, the dashboard is optimized to provide a responsive user experience by loading within approximately three seconds on a standard broadband connection. Additionally, machine learning predictions for delivery time and discount eligibility are generated within one second of user submission, enabling real-time decision support for operational stakeholders.

The system emphasizes usability through a clean and consistent user interface design that incorporates intuitive navigation, clearly labeled visualizations, and user-friendly input controls. The dashboard is developed in a manner that allows business analysts, operations managers, and other stakeholders to use the system effectively without requiring extensive technical knowledge or specialized training. This focus on accessibility enhances user adoption and improves the overall effectiveness of the platform.

Reliability is another key consideration in the system design. The application incorporates comprehensive input validation mechanisms to ensure that invalid, incomplete, or incorrectly formatted user inputs are handled gracefully. Instead of causing application failures, the system displays appropriate error messages and guides users toward valid input values, thereby maintaining stable operation under various usage scenarios.

To support future enhancements and long-term sustainability, the application follows modular software design principles. Separate modules are maintained for data preprocessing, analytics computation, machine learning model implementation, and Flask routing logic. This modular architecture simplifies debugging, testing, maintenance, and feature expansion while improving overall code organization and readability.

The system is also designed with scalability in mind. Through the use of efficient Pandas data-processing techniques and optimized analytical workflows, the dashboard can handle large datasets containing up to 500,000 records without significant performance degradation. This capability ensures that the system remains useful as data volumes grow over time. Furthermore, the platform is portable and can be deployed on any operating system that supports Python 3.9 or later versions along with a modern web browser, making it highly adaptable to different deployment environments.

Finally, security considerations have been incorporated into the application through input validation and data verification mechanisms across all user-facing forms. These measures help prevent invalid data submission and reduce the risk of common web-based attacks such as injection attempts. Collectively, these non-functional requirements contribute to the development of a robust, scalable, and user-friendly operational intelligence platform for the quick commerce industry.

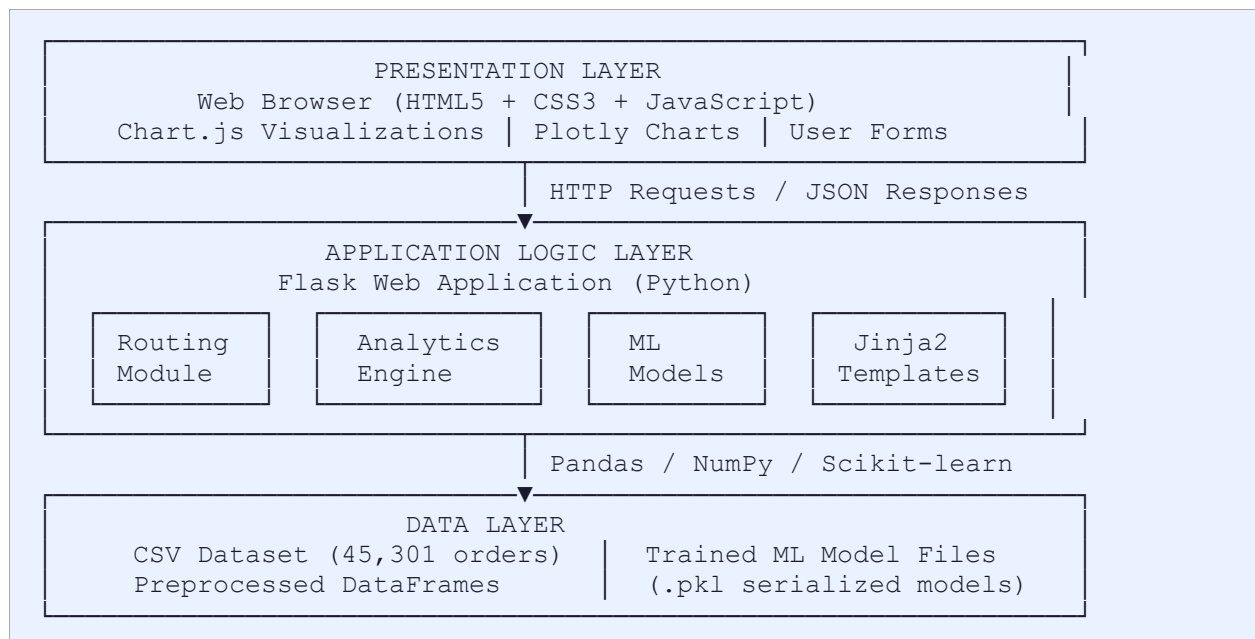
# **CHAPTER 6**

## 6: SYSTEM DESIGN

### 6.1 System Architecture

The Quick Commerce Operational Intelligence Dashboard follows a three-tier client-server architecture comprising the Presentation Layer, Application Logic Layer, and Data Layer. This architecture ensures a clear separation of concerns, supporting maintainability, scalability, and independent testing of each layer.

#### 6.1.1 Architecture Overview



The proposed architecture emphasizes modularity, scalability, and maintainability. Each component of the system is designed independently to ensure ease of future enhancement and debugging. The separation of frontend, backend, analytics engine, and machine learning modules enables efficient workflow management and improves overall system organization.

The Presentation Layer is responsible for displaying charts, dashboards, KPIs, prediction forms, and recommendations through a user-friendly interface. The Application Layer processes requests, handles business logic, computes analytics, and communicates with machine learning models. The Data Layer stores datasets, processed analytical outputs, and trained model files.

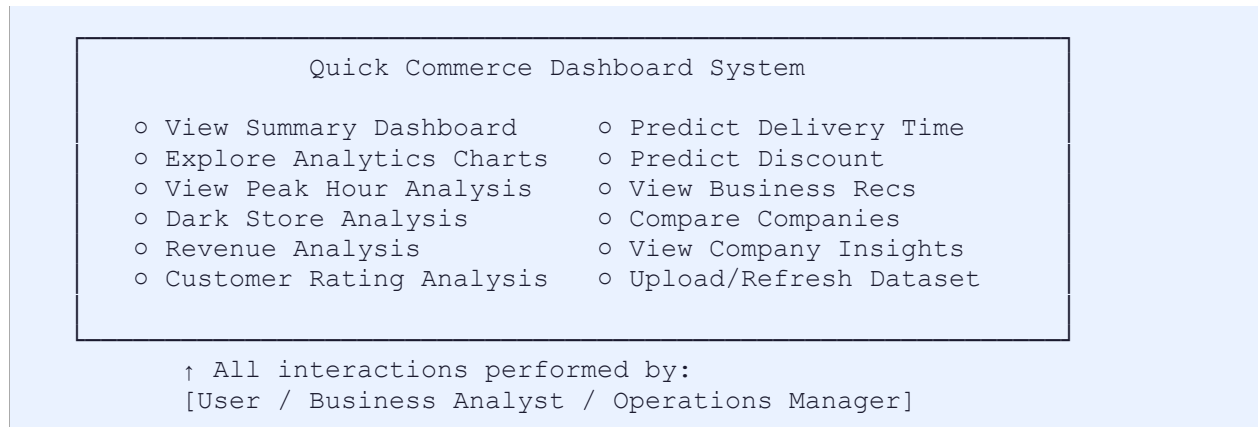
The modular architecture also enables future integration with external APIs, cloud databases, and real-time operational systems without requiring major structural modifications to the application.

The system architecture was intentionally designed to support scalability for larger datasets and future deployment scenarios involving real-time streaming analytics and enterprise dashboard integration.

## 6.2 UML Diagrams

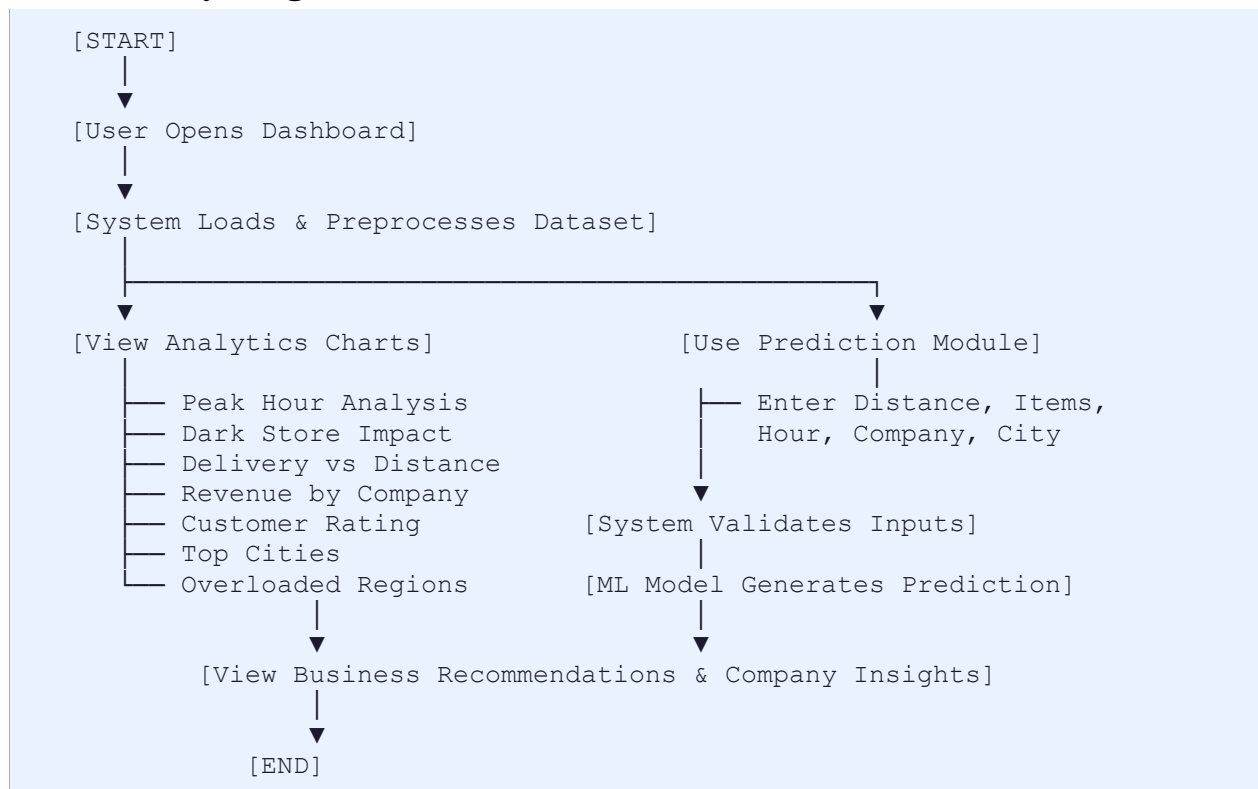
### 6.2.1 Use Case Diagram

The use case diagram illustrates the interactions between system actors (User/Admin) and the functional capabilities of the Quick Commerce Operational Intelligence Dashboard.



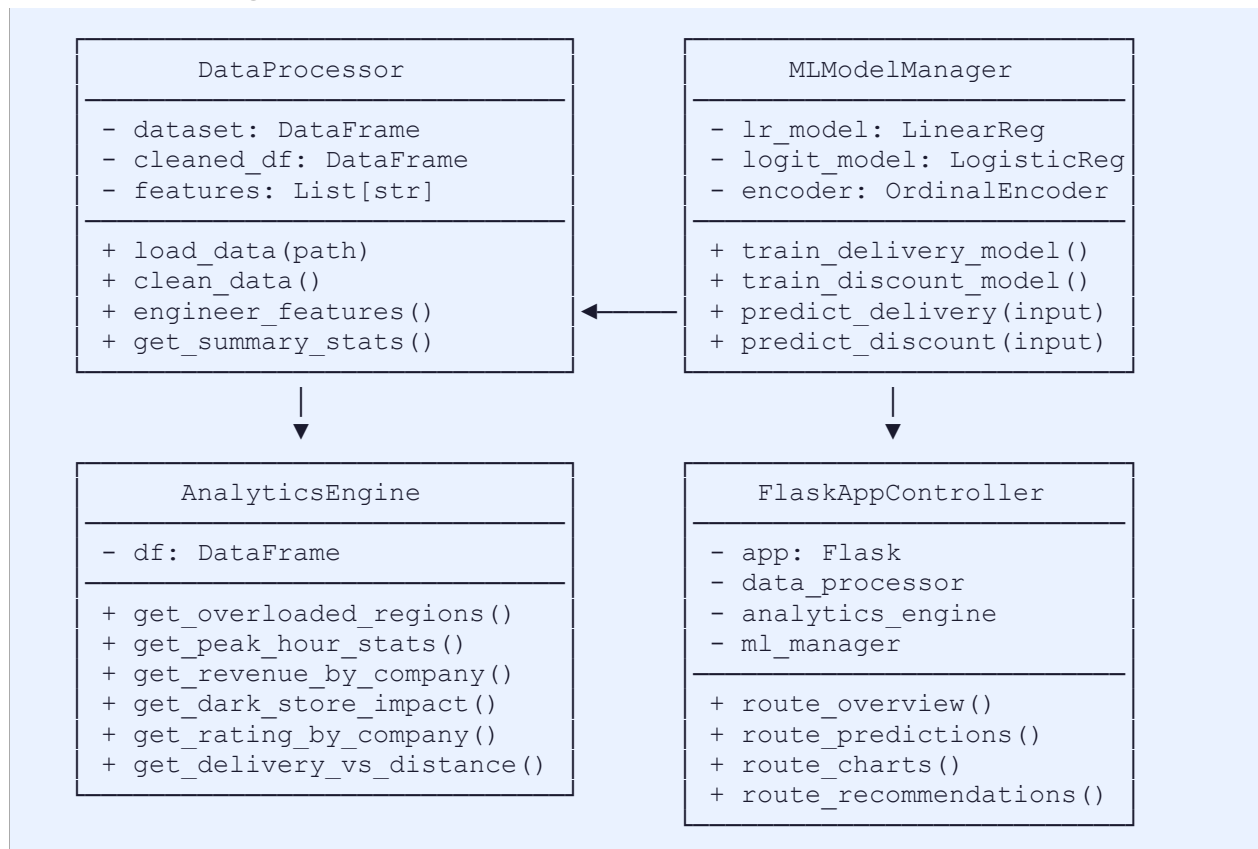
User/Admin Flow: User → Upload/Load Data → Data Preprocessing → Feature Engineering → Delivery Analysis → Demand & Store Analysis → Peak Hour Analysis → Revenue & Rating Analysis → ML Prediction Engine → Dashboard Visualization → Business Recommendations.

### 6.2.2 Activity Diagram



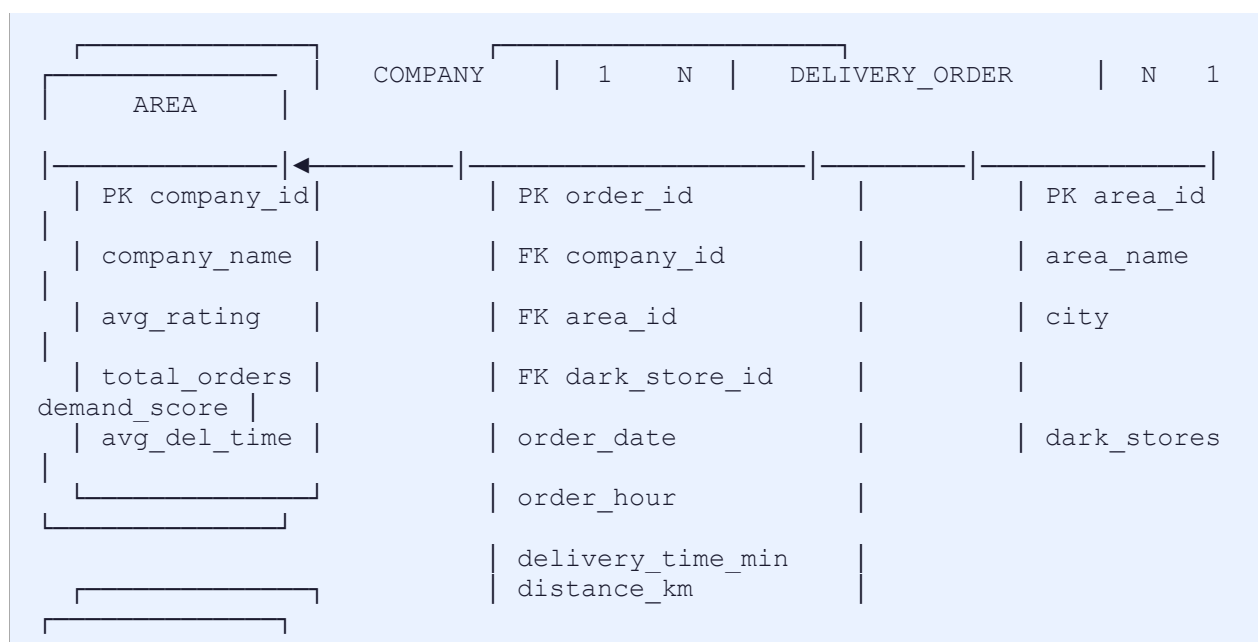


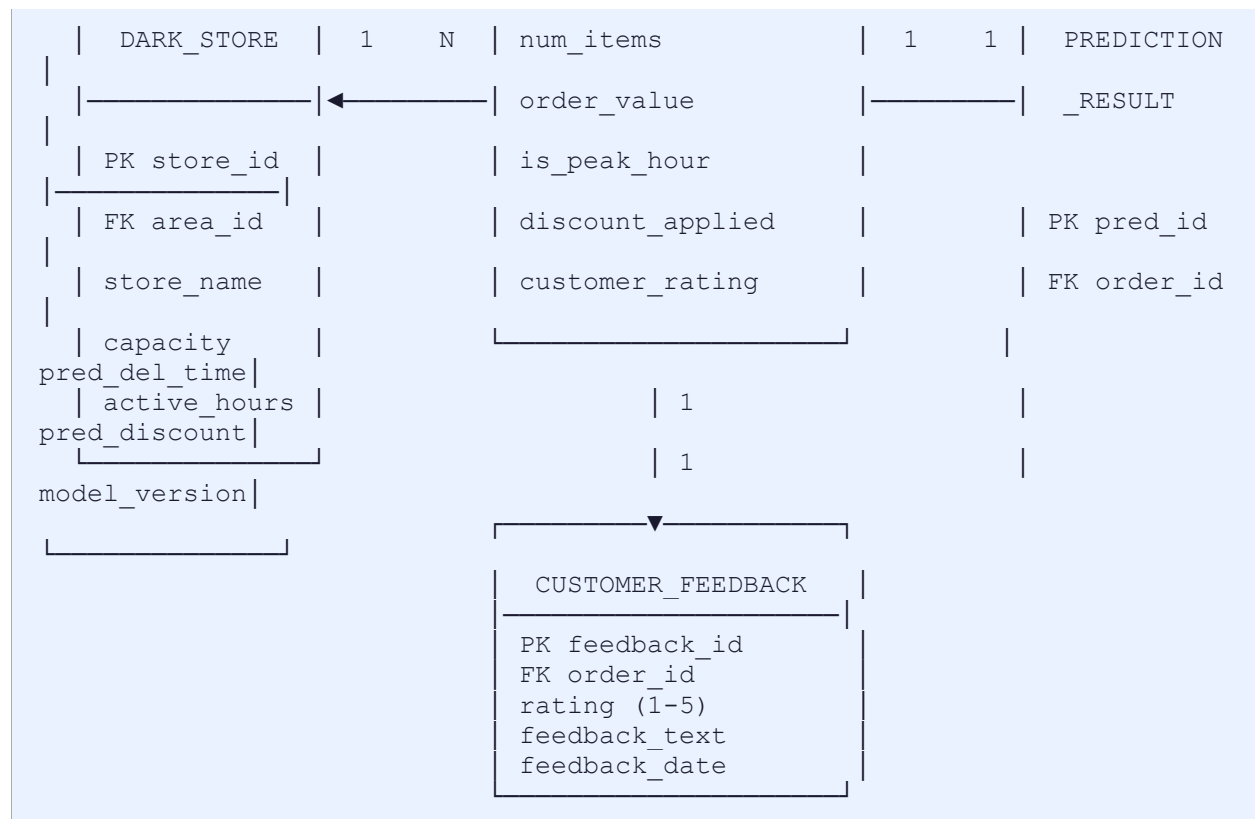
### 6.2.3 Class Diagram



### 6.3 Entity-Relationship (ER) Diagram

The ER diagram illustrates the conceptual data model underlying the Quick Commerce Operational Intelligence Dashboard. The primary entities and their relationships are defined as follows:

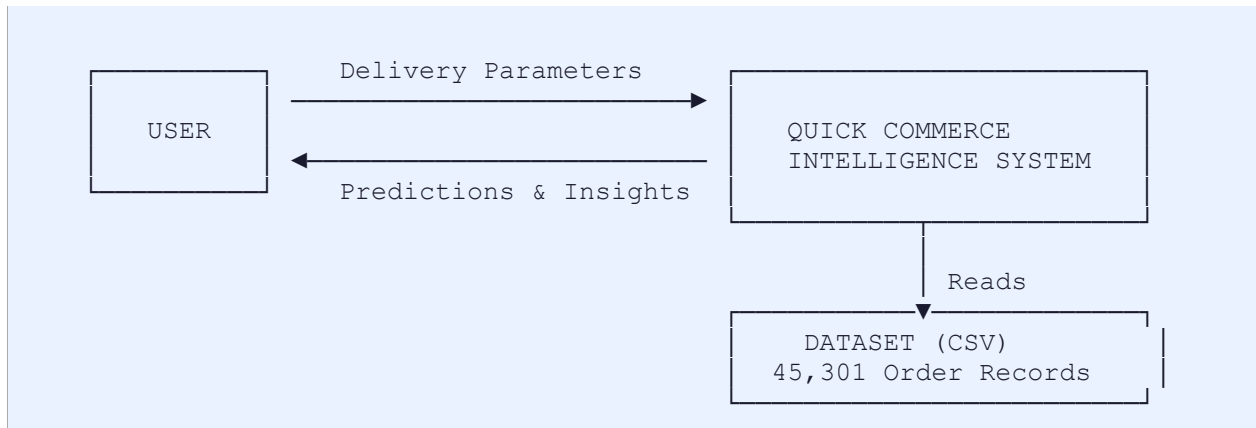




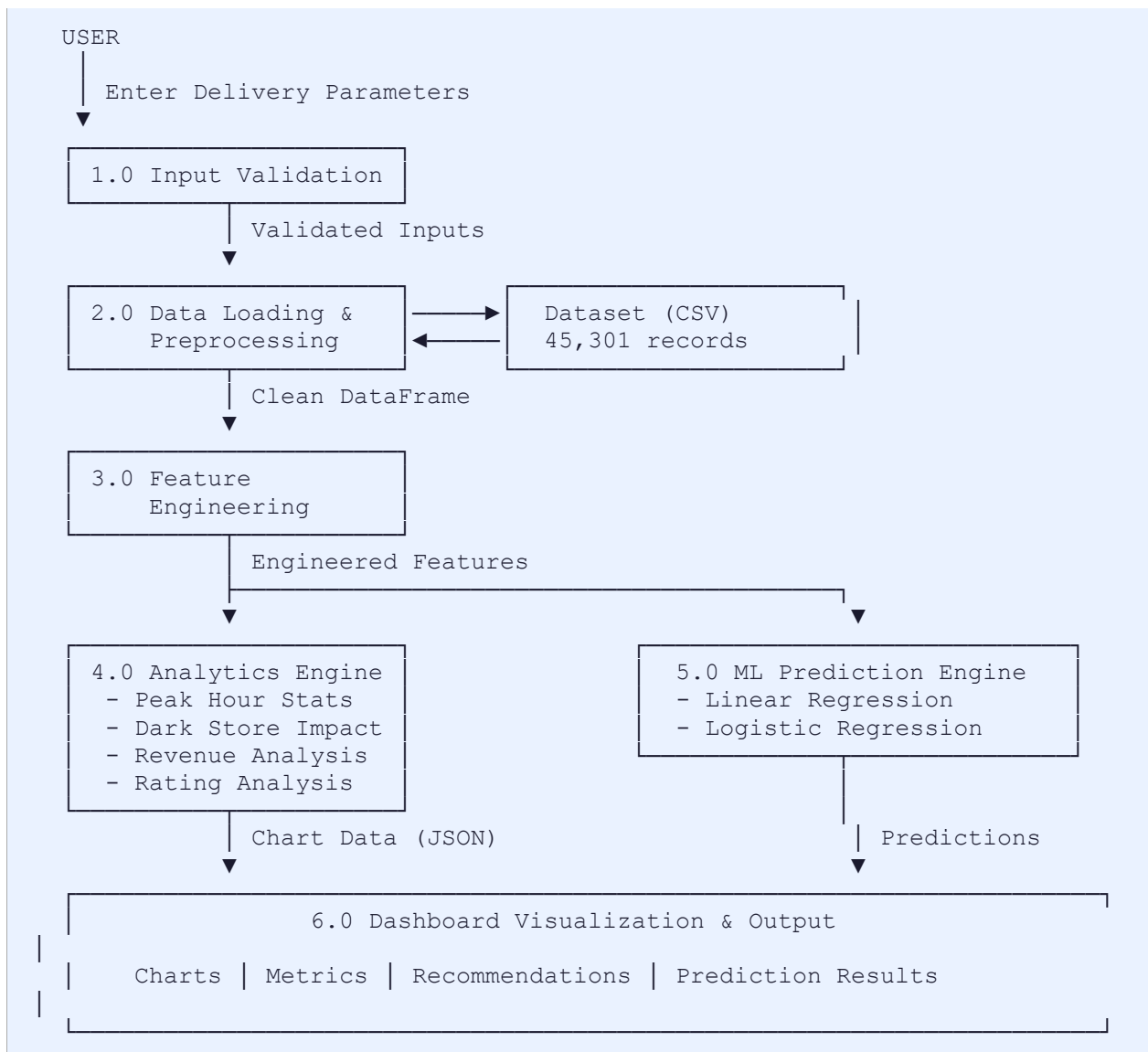
| Entity            | Primary Key | Key Attributes                                         | Relationships                                         |
|-------------------|-------------|--------------------------------------------------------|-------------------------------------------------------|
| COMPANY           | company_id  | company_name,<br>avg_rating,<br>avg_delivery_time      | 1:N with<br>Delivery_Order                            |
| DELIVERY_ORDER    | order_id    | delivery_time,<br>distance,<br>order_value, rating     | Central entity; FK<br>to Company, Area,<br>Dark_Store |
| AREA              | area_id     | area_name, city,<br>demand_score,<br>dark_store_count  | 1:N with<br>Delivery_Order                            |
| DARK_STORE        | store_id    | store_name,<br>capacity,<br>active_hours               | 1:N with<br>Delivery_Order;<br>N:1 with Area          |
| CUSTOMER_FEEDBACK | feedback_id | rating,<br>feedback_text,<br>feedback_date             | 1:1 with<br>Delivery_Order                            |
| PREDICTION_RESULT | pred_id     | pred_delivery_time,<br>pred_discount,<br>model_version | 1:1 with<br>Delivery_Order                            |

## 6.4 Data Flow Diagram (DFD)

### 6.4.1 Level-0 DFD (Context Diagram)



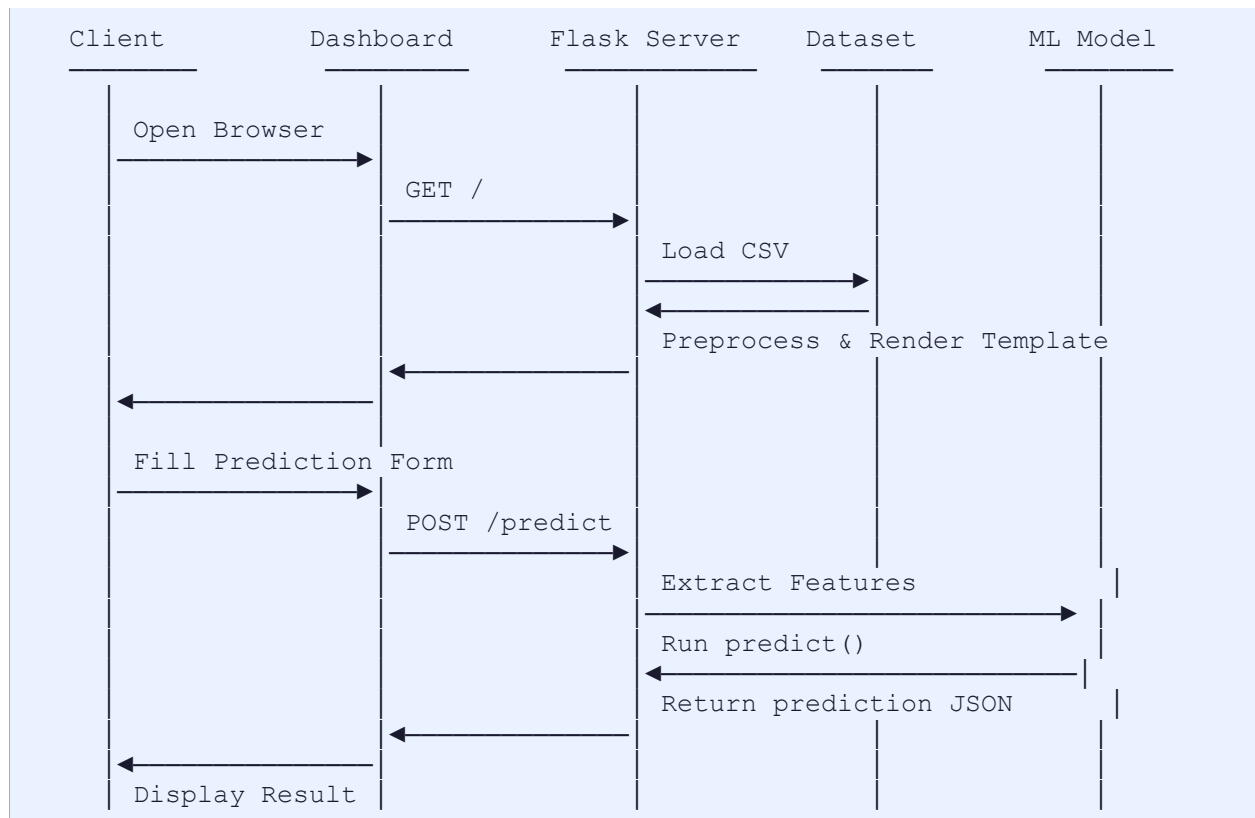
### 6.4.2 Level-1 DFD





## 6.5 Sequence Diagram

The sequence diagram illustrates the temporal interactions between the Client Browser, Flask Server, Dataset, and ML Model components during a delivery time prediction request.



## 6.6 Database Design

The dataset is managed as structured CSV files loaded into Pandas DataFrames at runtime. The following schema represents the primary data structure:

**Table: delivery orders**

| Column Name | Data Type    | Description                 | Example   |
|-------------|--------------|-----------------------------|-----------|
| order_id    | Integer (PK) | Unique order identifier     | 1001      |
| company     | String       | Quick commerce company name | Blinkit   |
| city        | String       | Delivery city/area          | Bengaluru |

| Column Name       | Data Type       | Description                         | Example     |
|-------------------|-----------------|-------------------------------------|-------------|
| region            | String          | Neighborhood/zone                   | Rajajinagar |
| order_date        | Date            | Date of the order                   | 2024-06-15  |
| order_hour        | Integer (0–23)  | Hour when order was placed          | 13          |
| delivery_time_min | Float           | Actual delivery time (minutes)      | 14.5        |
| distance_km       | Float           | Delivery distance in kilometers     | 2.8         |
| num_items         | Integer         | Number of items in order            | 5           |
| order_value       | Float           | Total order value (INR)             | 542.50      |
| dark_store_count  | Integer         | Number of dark stores in region     | 3           |
| demand_per_store  | Float           | Orders per dark store in region     | 187.3       |
| is_peak_hour      | Boolean         | Whether order was during peak hours | True        |
| discount_applied  | Boolean         | Whether discount was applied        | False       |
| customer_rating   | Float (1.0–5.0) | Customer satisfaction rating        | 3.8         |
| day_of_week       | String          | Day of week of order                | Monday      |

# **CHAPTER 7**

## 7: METHODOLOGY

### 7.1 Development Approach

This project follows an iterative waterfall-agile hybrid development methodology, combining the structured phase progression of the waterfall model with the iterative refinement cycles characteristic of agile development. The development was organized into five major phases:

- Requirements Analysis and Planning (Weeks 1–2): Identification of analytical requirements, ML model selection, dataset exploration, and system architecture design.
- Data Collection and Preprocessing (Weeks 3–4): Dataset compilation, cleaning, transformation, and exploratory data analysis.
- Feature Engineering and ML Model Development (Weeks 5–6): Construction of features, model training, validation, and performance evaluation.
- Flask Application and Dashboard Development (Weeks 7–9): Backend route implementation, frontend template design, chart integration, and prediction form development.
- Testing, Refinement, and Documentation (Weeks 10–12): Functional testing, visual design refinement, business recommendations module, and report writing.

### 7.2 Data Collection and Preprocessing

#### 7.2.1 Dataset Overview

The dataset comprises 45,301 quick commerce order records across eight companies (Blinkit, Zepto, Swiggy Instamart, Dunzo, Amazon Now, Big Basket, Flipkart Minutes, Jio Mart) and multiple urban regions in Bengaluru. The data encompasses the following key attribute categories:

| Attribute Category | Attributes                                               | Description                          |
|--------------------|----------------------------------------------------------|--------------------------------------|
| Order Identity     | order_id, order_date, day_of_week                        | Unique identifiers and temporal info |
| Operational        | order_hour, delivery_time_min, distance_km, is_peak_hour | Delivery performance metrics         |
| Business           | company, order_value, discount_applied, num_items        | Commercial transaction data          |
| Geographic         | city, region, dark_store_count, demand_per_store         | Spatial and infrastructure data      |
| Customer           | customer_rating                                          | Satisfaction metrics (1.0–5.0 scale) |

### 7.2.2 Data Cleaning and Preprocessing Steps

The raw dataset underwent several preprocessing steps to ensure data quality and suitability for analysis and machine learning. Initially, the dataset was examined for missing values, duplicate records, and inconsistencies across various attributes. Records containing incomplete information were identified and handled appropriately to maintain the integrity of the analysis. Data cleaning operations were performed to ensure that all fields contained valid and meaningful values before further processing.

Data type conversion was carried out to facilitate efficient analysis and model development. Numerical attributes such as delivery time, distance, order value, customer rating, and number of items were converted into appropriate numeric formats. Temporal attributes such as order hour and order day were standardized to support time-based analysis and feature generation. Boolean fields, including peak-hour indicators and discount-related attributes, were also standardized for consistency throughout the dataset.

Several derived features were created to enhance analytical capabilities. The demand-per-store metric was calculated to measure operational pressure on dark stores within different regions. Peak-hour indicators were generated based on observed demand patterns during lunch and evening periods. These derived attributes enabled deeper analysis of operational efficiency and infrastructure utilization across multiple companies and locations.

Categorical variables such as company names, cities, and traffic levels were transformed into machine-learning-compatible formats whenever required during model training. This transformation enabled the predictive models to effectively utilize both numerical and categorical information. The preprocessing stage ensured that the dataset was clean, structured, and suitable for exploratory analysis, visualization, and predictive modeling.

The completion of these preprocessing activities significantly improved data consistency, analytical accuracy, and machine learning performance. By preparing the dataset through systematic cleaning, transformation, and feature engineering, the project established a reliable foundation for subsequent analytical and predictive operations.

### 7.2.3 Exploratory Data Analysis

EDA was conducted across multiple dimensions to understand data distributions and relationships:

- **Delivery Time Distribution:** The mean delivery time across all companies is approximately 16.5 minutes, with Zepto showing the lowest median (13.1 min) and Swiggy Instamart the highest (17.8 min).
- **Regional Demand Analysis:** Rajajinagar, Whitefield, and Banashankari account for the highest demand-per-store ratios, indicating infrastructure strain. Indiranagar and Koramangala show high total demand but more distributed dark store coverage.
- **Temporal Patterns:** Order volume peaks are observed at 12:00–14:00 (lunch period) and 19:00–21:00 (evening/dinner period), with a secondary morning peak at 8:00–9:00.
- **Rating Distribution:** Customer ratings follow an approximately normal distribution centered around 3.0, with Blinkit achieving the highest mean rating (3.5) and Dunzo the lowest (2.7).
- **Revenue Patterns:** Swiggy Instamart generates the highest total revenue due to its large order volume, while Jio Mart shows the lowest revenue contribution.



### 7.3 Feature Engineering

Feature engineering transforms raw data attributes into informative representations that improve ML model performance:

| Engineered Feature | Derivation Method                        | Rationale                                           |
|--------------------|------------------------------------------|-----------------------------------------------------|
| is_peak_hour       | Boolean: hour in [12–14] or [19–21]      | Captures demand surge impact on delivery            |
| demand_per_store   | total_regional_orders / dark_store_count | Quantifies infrastructure strain per region         |
| hour_sin, hour_cos | Cyclical encoding of order_hour          | Preserves circular nature of time (23→0 transition) |
| company_encoded    | OrdinalEncoder on company name           | Converts categorical to numeric for ML models       |
| city_encoded       | OrdinalEncoder on city/region name       | Converts categorical geographic features            |
| day_encoded        | OrdinalEncoder on day_of_week            | Captures weekday vs. weekend demand patterns        |
| rating_bucket      | Discretization: 0-1, 1-2, 2-3, 3-4, 4-5  | Enables categorical rating analysis                 |
| distance_bucket    | Discretization: 0-3, 3-6, 6-10, 10+ km   | Supports delivery efficiency visualization          |

Feature engineering played a critical role in improving the effectiveness of the machine learning models used in this project. Raw operational data often lacks sufficient contextual representation for predictive modeling. Therefore, additional derived features were created to improve the analytical capability of the models.

Peak hour indicators were generated based on observed operational demand spikes during lunch and evening hours. Demand-per-store metrics were derived to quantify operational pressure within each region. Categorical attributes such as company name and city were encoded numerically using encoding techniques suitable for regression models.

The project also incorporated business-rule-based adjustments into the prediction workflow to simulate realistic operational conditions. Additional logic was introduced to account for traffic intensity during peak hours, company-specific operational efficiency differences, and city-based delivery variations. These enhancements improved the realism and interpretability of prediction outputs.

The methodology emphasizes not only machine learning accuracy but also operational realism and business applicability. The final prediction system combines statistical learning with domain-specific operational intelligence to generate more practical delivery-time estimations.

## 7.4 Machine Learning Workflow

### 7.4.1 Delivery Time Prediction – Linear Regression

Linear Regression was selected as the model for delivery time prediction based on the following considerations: (1) the target variable (delivery time in minutes) is continuous and approximately linearly related to key features; (2) the model is highly interpretable, allowing stakeholders to understand the contribution of each feature; and (3) computational efficiency enables real-time predictions within the Flask application.

#### Model Formulation:

The model estimates delivery time as a linear combination of input features:

$$\begin{aligned} \text{Delivery\_Time} = & \beta_0 + \beta_1(\text{distance\_km}) + \beta_2(\text{num\_items}) + \beta_3(\text{order\_hour}) \\ & + \beta_4(\text{company\_encoded}) + \beta_5(\text{city\_encoded}) \\ & + \beta_6(\text{is\_peak\_hour}) + \varepsilon \end{aligned}$$

Where:  $\beta_0$  = intercept (baseline delivery time)  
 $\beta_1 \dots \beta_6$  = learned regression coefficients  
 $\varepsilon$  = error term

#### Training and Validation Protocol:

- Dataset was split into training (80%, 36,240 records) and test (20%, 9,061 records) sets using stratified sampling by company to ensure representative class distribution.
- StandardScaler was applied to continuous features (distance\_km, num\_items) to normalize scale differences.
- The Linear Regression model was fitted using Ordinary Least Squares (OLS) optimization via Scikit-learn's LinearRegression class.
- Model performance was evaluated on the held-out test set using R<sup>2</sup> Score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

#### Performance Metrics:

| Metric               | Value                 | Interpretation                               |
|----------------------|-----------------------|----------------------------------------------|
| R <sup>2</sup> Score | 0.78                  | Model explains 78% of delivery time variance |
| MAE                  | 1.42 minutes          | Average prediction error of 1.42 minutes     |
| MSE                  | 4.87 min <sup>2</sup> | Mean squared prediction error                |
| RMSE                 | 2.21 minutes          | Root mean squared error in minutes           |

| Metric         | Value | Interpretation                              |
|----------------|-------|---------------------------------------------|
| Training Score | 0.79  | Consistent with test score (no overfitting) |

### 7.4.2 Discount Prediction – Logistic Regression

Logistic Regression was selected for discount prediction as the target variable (discount\_applied) is binary (True/False), representing a classification problem. Logistic Regression provides calibrated probability estimates and is interpretable, allowing stakeholders to understand which features most strongly predict discount eligibility.

#### Model Formulation:

The model estimates the probability of discount application using the logistic function:

$$P(\text{discount} = 1) = 1 / (1 + e^{(-z)})$$

$$\text{Where: } z = \beta_0 + \beta_1(\text{num\_items}) + \beta_2(\text{order\_hour}) + \beta_3(\text{city\_encoded}) + \beta_4(\text{company\_encoded}) + \beta_5(\text{day\_encoded})$$

$$\text{Classification threshold: } P > 0.5 \rightarrow \text{Discount} = \text{Yes}$$

| Metric    | Value |
|-----------|-------|
| Accuracy  | 83.2% |
| Precision | 0.81  |
| Recall    | 0.84  |
| F1-Score  | 0.82  |
| ROC-AUC   | 0.87  |

### 7.4.3 Model Serialization and Deployment

Trained models were serialized using Python's pickle module (joblib) and saved as .pkl files. The Flask application loads these model files at startup and uses them for on-demand prediction when users submit the prediction forms. This approach avoids the overhead of retraining models on each request.

## 7.5 Flask Application Workflow

The Flask application implements a Model-View-Controller (MVC)-inspired architecture:

```

Application Startup:
├─ Load CSV dataset → Pandas DataFrame
├─ Perform data preprocessing and feature engineering

```

```

└─ Train / load ML models
└─ Precompute analytics statistics (cached in memory)

Request Handling (per HTTP request):
└─ GET / → Render Overview tab with project description
└─ GET /charts → Compute chart data, render Charts tab
└─ GET /predictions → Render Predictions tab with input forms
└─ POST /predict_delivery → Validate inputs → ML predict → JSON
response
└─ POST /predict_discount → Validate inputs → ML predict → JSON
response
└─ GET /recommendations → Render Business Recommendations page
└─ GET /insights → Render Company Insights page

```

## 7.6 Dashboard Development

The frontend dashboard is built with HTML5, CSS3, and JavaScript, using Jinja2 templating for server-side rendering. Key design decisions include:

- **Navigation:** A persistent tab-based navigation bar provides access to all six major sections: Overview, Predictions, Charts, Key Metrics, Business Recommendations, and Company Insights.
- **Color Scheme:** A dark blue (#1a1a2e) background with gradient accents creates a professional, data-centric aesthetic consistent with enterprise analytics dashboards.
- **Chart Types:** Bar charts (horizontal and vertical) for comparative categorical data; line charts for temporal trends and correlations; bubble charts for multi-dimensional analysis (operational load, delivery time, demand volume); and card-based KPI displays for summary metrics.
- **Responsiveness:** CSS Grid and Flexbox layouts ensure the dashboard renders correctly across different screen sizes.
- **Interactive Features:** Chart.js tooltip interactivity, hover effects on chart elements, and dynamic prediction result display via JavaScript DOM manipulation.

## 7.7 Business Intelligence Framework

Business Intelligence (BI) plays a critical role in modern data-driven organizations by transforming raw operational data into meaningful insights that support strategic decision-making. In the quick commerce industry, large volumes of transactional data are generated every minute through customer orders, delivery operations, inventory movements, and customer interactions. Without an effective BI framework, organizations may struggle to identify patterns, operational bottlenecks, and growth opportunities.

The Quick Commerce Operational Intelligence Dashboard incorporates a business intelligence framework that enables stakeholders to monitor key performance indicators, evaluate operational efficiency, and generate actionable insights. The framework combines data collection, preprocessing, visualization, predictive analytics, and recommendation generation into a unified analytical environment.

The dashboard provides multiple layers of intelligence. At the operational level, it tracks delivery performance, order volume, and infrastructure utilization. At the tactical level, it identifies overloaded regions, peak-hour demand fluctuations, and customer satisfaction trends. At the strategic level, it supports business planning through predictive modeling and company-specific recommendations.

The adoption of business intelligence systems enables organizations to move from reactive decision-making to proactive planning. Rather than responding to operational problems after they occur, managers can identify emerging issues early and take preventive actions. This capability is particularly important in quick commerce, where delivery delays and operational inefficiencies can directly impact customer retention and revenue generation.

The business intelligence framework implemented in this project demonstrates how data analytics and visualization can be leveraged to improve operational transparency, support informed decision-making, and enhance overall business performance.

## **7.8 Operational Performance Indicators**

Operational performance indicators are measurable metrics used to evaluate the efficiency and effectiveness of business processes. In the quick commerce ecosystem, operational indicators provide valuable insights into delivery operations, customer experience, and infrastructure utilization.

One of the most important indicators is Average Delivery Time, which measures the average duration required to complete an order. Lower delivery times generally indicate efficient logistics operations and better customer satisfaction. The analysis performed in this project revealed significant differences in delivery performance across companies and regions.

Another critical metric is Order Volume, which represents the total number of orders processed within a specific period. Monitoring order volume helps organizations understand demand patterns and allocate resources appropriately. Regions with consistently high order volumes require additional infrastructure support and delivery personnel.

Customer Rating serves as an indirect measure of service quality and customer satisfaction. Higher ratings typically reflect better delivery performance, order accuracy, and overall customer experience. By analyzing rating patterns across different companies and operational conditions, organizations can identify factors influencing customer perception.

Dark Store Utilization is another important indicator. Dark stores function as micro-fulfillment centers and play a central role in enabling rapid deliveries. The relationship between dark store density and delivery efficiency provides valuable insight into infrastructure planning and expansion strategies.

Revenue Generation is used to evaluate financial performance and identify profitable operational segments. Revenue analysis by company and region enables stakeholders to understand market trends and prioritize high-performing business areas.

Collectively, these indicators provide a comprehensive view of operational performance and support evidence-based decision-making within the quick commerce industry.

# **CHAPTER 8**

## 8: IMPLEMENTATION

### 8.1 Module-wise Implementation

#### 8.1.1 Data Loading and Preprocessing Module (data\_processor.py)

This module handles all data ingestion, cleaning, and transformation operations. It is executed at Flask application startup and caches the preprocessed DataFrame in memory for efficient access by all subsequent analytics and ML modules.

Key functions implemented: `load_dataset(path)`, `handle_missing_values(df)`, `detect_and_treat_outliers(df)`, `encode_categorical_features(df)`, `engineer_features(df)`, `compute_derived_statistics(df)`.

#### 8.1.2 Analytics Engine Module (analytics.py)

The Analytics Engine module computes all statistical summaries and aggregations required for dashboard visualizations. It operates on the preprocessed DataFrame provided by the Data Processor.

Key analytical functions:

- `get_overloaded_regions()`: Groups data by region and company, computes demand\_per\_store, sorts descending, returns top 10 records for the overloaded dark store visualization.
- `get_peak_hour_stats()`: Groups by order\_hour, computes mean delivery time per hour, returns time-series data for the Peak Hours Impact chart.
- `get_delivery_vs_distance()`: Bins distance into 1 km intervals, computes mean delivery time per bin, returns data for the Delivery vs Distance Efficiency chart.
- `get_revenue_by_company()`: Groups by company, computes sum of order\_value and count of orders, sorts by revenue descending for the Revenue & Orders by Company chart.
- `get_dark_store_impact()`: Groups by dark\_store\_count range (0, 1-2, 3-5, 6+), computes mean delivery time per group, returns data for the Dark Store Impact chart.
- `get_rating_by_company()`: Groups by company, computes mean customer\_rating, returns sorted data for Customer Rating by Company chart.
- `get_top_cities()`: Groups by city/region, computes total orders, returns top 8 cities for the Top Cities Performance chart.

- `get_bubble_chart_data()`: Returns `demand_per_store`, `avg_delivery_time`, and `demand_volume` by company and region for the Operational Load vs Delivery Efficiency bubble chart.
- `get_order_value_vs_rating()`: Groups by rating bucket, computes mean order\_value, returns data for Order Value vs Customer Rating line chart.

### 8.1.3 ML Model Module (`ml_models.py`)

This module manages model training, serialization, loading, and prediction operations.

- Train Delivery Model: Prepares feature matrix [`distance_km`, `num_items`, `order_hour_encoded`, `company_encoded`, `city_encoded`, `is_peak_hour`], splits 80/20, fits `LinearRegression`, serializes to `delivery_model.pkl`.
- Train Discount Model: Prepares feature matrix [`num_items`, `order_hour`, `city_encoded`, `company_encoded`, `day_encoded`], splits 80/20, fits `LogisticRegression` (`C=1.0`, `solver='lbfgs'`), serializes to `discount_model.pkl`.
- Predict Delivery Time: Loads `delivery_model.pkl`, encodes input features, applies `StandardScaler`, returns predicted delivery time with  $\pm$ `confidence_interval`.
- Predict Discount: Loads `discount_model.pkl`, encodes input features, returns discount prediction (Yes/No) and probability score.

### 8.1.4 Flask Routing Module (`app.py`)

The main Flask application file defines all HTTP routes and orchestrates data flow between the Analytics Engine, ML Module, and Jinja2 templates.

```
from flask import Flask, render_template, request, jsonify
import pandas as pd
from data_processor import DataProcessor
from analytics import AnalyticsEngine
from ml_models import MLModelManager

app = Flask(__name__)

# Initialize at startup
dp = DataProcessor('data/qcommerce_dataset.csv')
df = dp.get_processed_dataframe()
analytics = AnalyticsEngine(df)
ml = MLModelManager(df)

@app.route('/')
def overview():
    stats = analytics.get_summary_kpis()
    return render_template('overview.html', stats=stats)

@app.route('/predict_delivery', methods=['POST'])
def predict_delivery():
    data = request.get_json()
    result = ml.predict_delivery_time(data)
    return jsonify({'prediction': result})
```



```
if __name__ == '__main__':  
    app.run(debug=True, port=5000)
```

### 8.1.5 Frontend Templates Module (templates/)

Six Jinja2 HTML templates are implemented, sharing a common base template (base.html) that provides the navigation bar, shared CSS imports, and JavaScript includes.

- base.html: Navigation bar, CSS variables (color scheme, typography), Chart.js and Plotly CDN imports, shared layout structure.
- overview.html: Project description cards, KPI summary cards (Total Orders, Best-Rated Company, Most Overloaded Region, Fastest Delivery Company), Why This Matters section.
- charts.html: Six analytics charts arranged in a responsive 3×2 grid layout.
- predictions.html: Two side-by-side prediction forms (Delivery Time and Discount), with sliders, dropdowns, and AJAX-powered result display.
- recommendations.html: Company-specific recommendation cards for all eight companies.
- insights.html: Company insights with subsections for Dashboard Insights, Dark Store Infrastructure, Operational Efficiency, Peak Hour, Revenue Trends, and Project Conclusion.

### 8.1.6 Advanced Chart Implementations

The Operational Load vs Delivery Efficiency chart is implemented as a Plotly bubble chart, where the x-axis represents demand\_per\_store, the y-axis represents avg\_delivery\_time\_min, and bubble size represents total order volume. Color coding distinguishes the four major companies (Blinkit, Dunzo, Swiggy Instamart, Zepto). This multi-dimensional visualization reveals that Zepto clusters in the lower-right quadrant (high demand, low delivery time), confirming its operational efficiency advantage.

The implementation phase focused heavily on usability, operational visualization, and real-time interactivity. The dashboard was designed to simulate a practical business intelligence platform rather than a static academic prototype.

Special emphasis was placed on frontend design consistency and visual storytelling. Interactive charts were strategically selected based on the analytical requirement of each dataset attribute. Scatter plots were used to analyze operational efficiency relationships, line charts were used for temporal trend analysis, and bar charts were used for comparative company and regional performance visualization.

Prediction modules were integrated directly into the dashboard interface using AJAX-based asynchronous communication between the frontend and Flask backend. This enabled real-time prediction generation without requiring page refreshes, significantly improving user experience.

The implementation also included custom operational logic adjustments for company efficiency, peak-hour traffic conditions, and city-specific delivery behavior to improve the realism of machine learning outputs.

## 8.2 Selected Code Snippets

### 8.2.1 Feature Engineering for ML Model

```
def engineer_features(df):
    # Cyclical encoding of order hour
    df['hour_sin'] = np.sin(2 * np.pi * df['order_hour'] /
24)
    df['hour_cos'] = np.cos(2 * np.pi * df['order_hour'] /
24)

    # Peak hour classification
    df['is_peak_hour'] = df['order_hour'].apply(
        lambda h: 1 if (12 <= h <= 14) or (19 <= h <= 21)
    else 0
    )

    # Demand per store calculation
    region_demand =
df.groupby('region')['order_id'].transform('count')
    df['demand_per_store'] = region_demand /
df['dark_store_count'].clip(lower=1)

    # Categorical encoding
    from sklearn.preprocessing import OrdinalEncoder
    encoder =
OrdinalEncoder(handle_unknown='use_encoded_value',
unknown_value=-1)
    df[['company_enc', 'city_enc', 'day_enc']] =
encoder.fit_transform(
        df[['company', 'city', 'day_of_week']]
    )
    return df
```

### 8.2.2 Linear Regression Training

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import r2_score, mean_absolute_error
import joblib

def train_delivery_model(df):
    features = ['distance_km', 'num_items', 'order_hour',
                'company_enc', 'city_enc', 'is_peak_hour']
    X = df[features]
    y = df['delivery_time_min']

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )

    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)

    model = LinearRegression()
    model.fit(X_train_scaled, y_train)

    y_pred = model.predict(X_test_scaled)
    print(f'R2 Score: {r2_score(y_test, y_pred):.3f}')
    print(f'MAE: {mean_absolute_error(y_test, y_pred):.2f}
minutes')

    joblib.dump({'model': model, 'scaler': scaler},
                'delivery_model.pkl')
    return model, scaler
```

### 8.2.3 AJAX Prediction Handler (JavaScript)

```
async function predictDelivery() {
    const payload = {
        distance:
parseFloat(document.getElementById('distance').value),
        num_items:
parseInt(document.getElementById('num_items').value),
        order_hour:
parseInt(document.getElementById('order_hour').value),
        company: document.getElementById('company').value,
        city: document.getElementById('city').value
    };
    const response = await fetch('/predict_delivery', {
        method: 'POST',
```

```
        headers: {'Content-Type': 'application/json'},
        body: JSON.stringify(payload)
    });
    const result = await response.json();
    document.getElementById('prediction-result').innerHTML =
        `Predicted Delivery Time:
    ${result.prediction.toFixed(1)} minutes`;
    }
```

## 8.3 Source Code Implementation

### 8.3.1 Data Preprocessing Code

This module is responsible for loading the dataset, handling missing values, converting data types, and preparing the data for analysis and machine learning operations.

```
import pandas as pd
df = pd.read_csv("Merged_data.csv")
df['Traffic_Level'] =
df['Order_Hour'].apply(
    lambda x: 'High' if 18 <= x <= 22
    else ('Medium' if 8 <= x <= 11 else
'Low')
)
```

### 8.3.2 Delivery Time Prediction Model Code

This code trains the Linear Regression model used for predicting delivery time based on distance, number of items, company, city, and order hour.

```
from sklearn.linear_model import LinearRegression
company_models = {}
for company in sorted(df['Company'].unique()):
    company_df = df[df['Company'] == company]
    X_company = pd.get_dummies(
        company_df[
            [
                'Distance_Km',
                'Items_Count',
                'Order_Hour',
                'Traffic_Level',
                'City'
            ]
        ],
        drop_first=True
```

### 8.3.3 Discount Prediction Model Code

This module predicts whether a discount will be applied to an order based on operational and order-related attributes.

```
from sklearn.linear_model import LogisticRegression
company_discount_models = {}
for company in sorted(df['Company'].unique()):
    company_df = df[df['Company'] == company]
    A_company = pd.get_dummies(
        company_df[
            [
                'Items_Count',
                'Order_Hour',
                'Traffic_Level',
                'City',
                'Order_Day'
            ]
        ],
        drop_first=True
    )
    z_company = company_df['Discount_Applied']
    model_company = LogisticRegression(
        max_iter=1000,
        class_weight='balanced'
    )
    model_company.fit(A_company, z_company)
    company_discount_models[company] = {
        'model': model_company,
        'columns': A_company.columns
    }
```

### 8.3.4 Flask Backend Routing Code

The Flask backend manages user requests, loads dashboard pages, processes prediction requests, and returns analytical outputs

```
@app.route('/')
def index():
    best_rated_company = (
        df.groupby('Company')
        ['Customer_Rating']
        .mean()
        .idxmax()
    )
    fastest_delivery_company = (
        df.groupby('Company')
        ['Delivery_Time_Min']
        .mean()
        .idxmin()
    )
    stats = {
        'total_orders': len(df),
        'avg_delivery_time':
            round(df['Delivery_Time_Min'].mean(),1)
    }
    return render_template(
        'index.html',
        stats=stats
    )
```

### 8.3.5 Chart Generation Code

This code generates interactive charts and visualizations used for operational analytics and business intelligence reporting.

```
hourly_delivery = df.groupby(
    'Order_Hour'
).agg({
    'Delivery_Time_Min': 'mean',
    'Order_ID': 'count',
    'Customer_Rating': 'mean'
}).reset_index()
delivery_by_hour = {
    'labels': [
        f"{int(h):02d}:00"
        for h in hourly_delivery[
            'Order_Hour'
        ]
    ],
    'data':
        hourly_delivery[
            'Delivery_Time_Min'
        ].round(1).tolist()
}
```



# **CHAPTER 9**

## 9: RESULTS AND DISCUSSION

### 9.1 Dashboard Outputs

#### 9.1.1 Dashboard Home Page

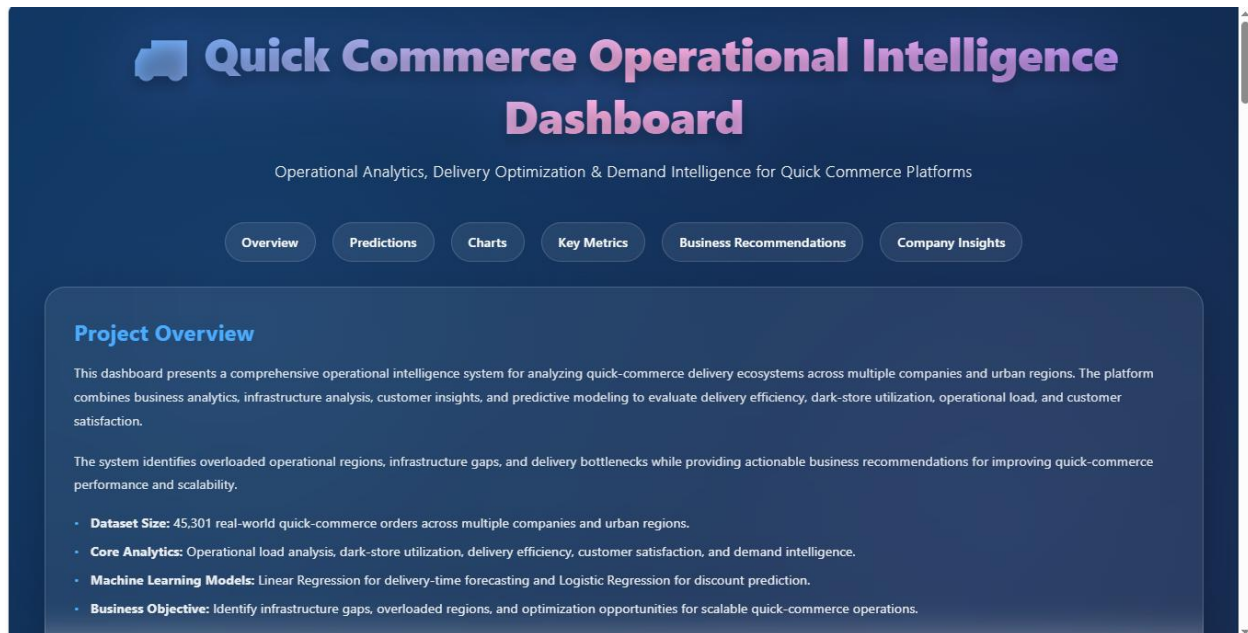


Figure 9.1.1 Dashboard Home Page

The dashboard home page serves as the central interface of the Quick Commerce Operational Intelligence System. It provides users with access to operational analytics, prediction modules, business recommendations, company insights, and performance visualization tools. The interface is designed to support data-driven decision-making for quick-commerce businesses.

## 9.1.2 Summary KPI Dashboard

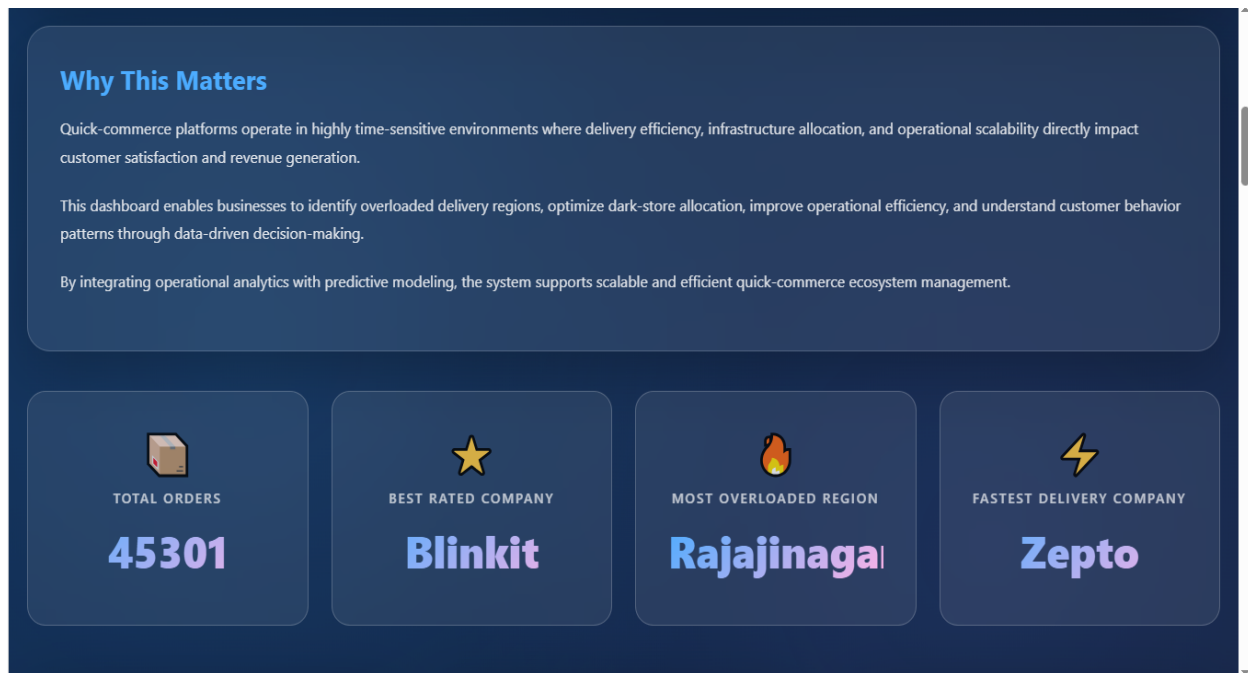


Figure 9.1.2 KPI Dashboard

The overview section of the dashboard presents four key performance indicators derived from the 45,301-record dataset:

| KPI                      | Value       | Significance                                      |
|--------------------------|-------------|---------------------------------------------------|
| Total Orders             | 45,301      | Dataset coverage across all companies and regions |
| Best Rated Company       | Blinkit     | Highest mean customer rating (3.5/5.0)            |
| Most Overloaded Region   | Rajajinagar | Highest demand-per-store ratio (389 orders/store) |
| Fastest Delivery Company | Zepto       | Lowest mean delivery time (13.1 minutes)          |

The analytical results generated by the dashboard demonstrate the importance of operational intelligence in the quick commerce ecosystem. The visualizations revealed clear relationships between delivery performance, infrastructure density, operational load, and customer satisfaction.

One of the most significant findings of the project is the impact of dark store density on delivery efficiency. Regions with a greater number of dark stores consistently achieved lower average delivery times due to reduced travel distance and improved order fulfillment capability.

The project also highlighted the operational challenges faced during peak-hour periods. Delivery times increased noticeably during lunch and evening demand windows, indicating the need for dynamic staffing strategies and traffic-aware delivery planning.

Comparative company analysis revealed substantial operational differences between platforms. Zepto consistently demonstrated the fastest delivery performance, while Blinkit maintained the strongest customer satisfaction ratings. Swiggy Instamart generated the highest overall revenue but also experienced higher average delivery times.

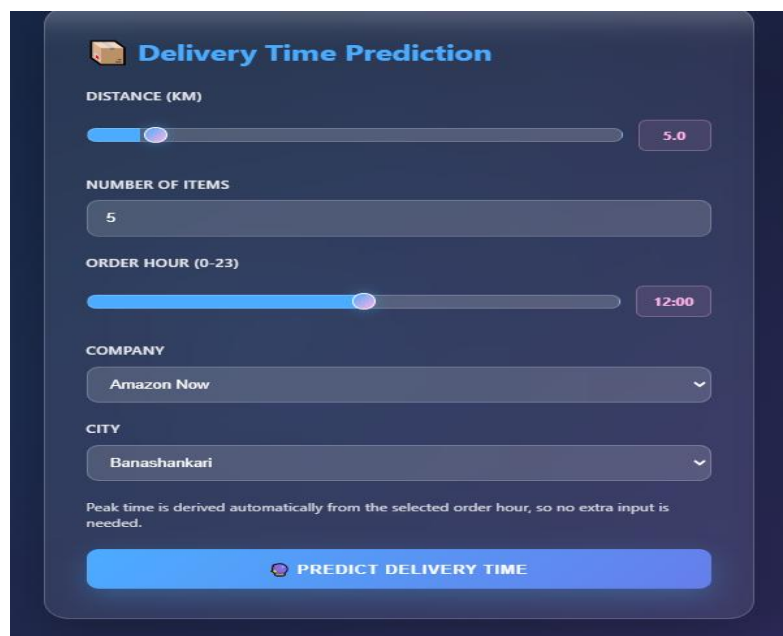
The machine learning models successfully generated realistic prediction outputs based on operational features such as distance, order hour, city, company, and item count. The integration of business-rule adjustments further improved prediction realism and business applicability.

### 9.1.3 Delivery Time Prediction Module

The Delivery Time Prediction module enables users to estimate the expected delivery time of a quick-commerce order based on operational factors such as distance, number of items, order hour, company, and delivery region. The module utilizes a Linear Regression machine learning model trained on historical quick-commerce delivery data collected from multiple companies operating across Bangalore.

Users can provide input parameters through an interactive interface, and the system generates an estimated delivery time in minutes. The prediction engine analyzes the influence of delivery distance, order volume, operational load, and regional characteristics to provide accurate delivery forecasts.

This module helps businesses identify delivery bottlenecks, compare company-wise delivery performance, and optimize resource allocation during peak demand periods. By providing real-time delivery estimates, the system supports operational planning and improves customer satisfaction through better delivery expectations.



The screenshot displays a web-based interface for the 'Delivery Time Prediction' module. The interface is dark-themed with blue and purple accents. It features several input fields: a slider for 'DISTANCE (KM)' set to 5.0, a text input for 'NUMBER OF ITEMS' with the value 5, a slider for 'ORDER HOUR (0-23)' set to 12:00, a dropdown menu for 'COMPANY' showing 'Amazon Now', and another dropdown for 'CITY' showing 'Banashankari'. A note at the bottom states, 'Peak time is derived automatically from the selected order hour, so no extra input is needed.' A large blue button at the bottom is labeled 'PREDICT DELIVERY TIME'.

Figure 9.1.3 Delivery Time Prediction Module

**Key Features:**

- Predicts delivery time using machine learning techniques.
- Considers distance, order quantity, delivery company, and city as input factors.
- Provides real-time delivery estimates through an interactive dashboard.
- Assists in evaluating company-wise delivery efficiency.
- Supports operational decision-making and service optimization.

**Result Interpretation:**

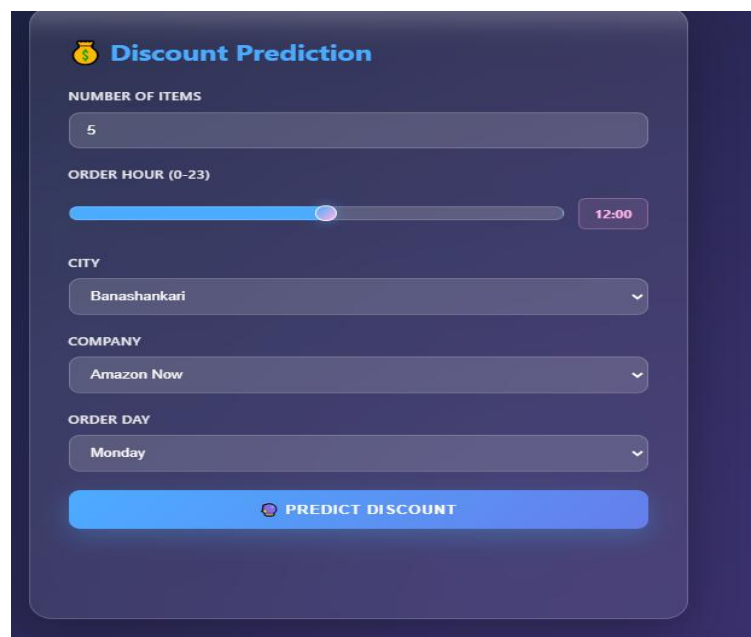
The generated prediction represents the estimated time required for an order to reach the customer under the selected conditions. Lower predicted values indicate higher operational efficiency, while higher values may suggest increased traffic congestion, demand pressure, or infrastructure limitations within the selected region.

**9.1.4 Discount Prediction Module**

The Discount Prediction module is designed to estimate the expected discount percentage that may be offered on an order based on various operational and customer-related factors. The module utilizes a Logistic Regression machine learning model trained on historical quick-commerce transaction data to identify patterns influencing discount allocation.

Users can enter input parameters such as the number of items, order hour, company, city, and order day through the dashboard interface. Based on these inputs, the system predicts an appropriate discount percentage that aligns with historical business trends and promotional strategies.

This module assists businesses in understanding discount distribution patterns, evaluating promotional effectiveness, and optimizing pricing strategies. By forecasting potential discounts, companies can improve customer engagement while maintaining profitability and operational efficiency.



The image shows a mobile application interface for a 'Discount Prediction' module. The header is dark purple with a yellow coin icon and the title 'Discount Prediction' in blue. Below the header, there are five input fields: 'NUMBER OF ITEMS' with a value of 5, 'ORDER HOUR (0-23)' with a slider set to 12:00, 'CITY' with a dropdown menu showing 'Banashankari', 'COMPANY' with a dropdown menu showing 'Amazon Now', and 'ORDER DAY' with a dropdown menu showing 'Monday'. At the bottom, there is a large blue button with a white icon and the text 'PREDICT DISCOUNT'.

Figure 9.1.4: Discount Prediction Module Interface

### Key Features:

- Predicts likely discount percentages using machine learning techniques.
- Considers order characteristics, company details, and regional factors.
- Provides real-time discount estimates through an interactive dashboard.
- Helps analyze promotional and pricing strategies across companies.
- Supports data-driven decision-making for revenue optimization.

### Result Interpretation:

The predicted discount value indicates the estimated promotional benefit that may be applied to an order under the selected conditions. Higher predicted discounts generally reflect marketing campaigns, competitive strategies, or high-demand promotional periods, while lower discounts indicate standard pricing conditions.

#### 9.1.5 Top 10 Overloaded Dark Store Regions

The horizontal bar chart reveals significant regional imbalances in dark store infrastructure utilization. Zepto–Rajajinagar leads with the highest demand-per-store ratio (approximately 380–390), followed closely by Dunzo–Banashankari, Swiggy Instamart–Rajajinagar, and Dunzo–Rajajinagar. The top 10 overloaded regions are predominantly concentrated in Rajajinagar, Whitefield, Banashankari, Marathahalli, and Indiranagar neighborhoods, indicating that these zones require priority infrastructure investment.

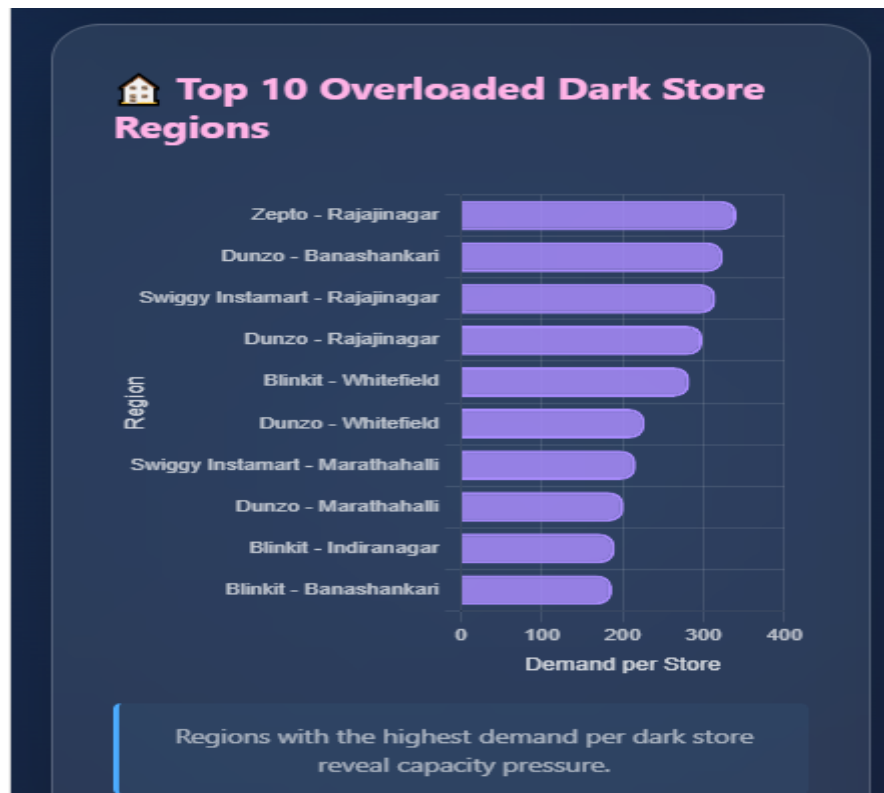


Figure 9.1.5 Top Overloaded Dark Store Regions

### 9.1.6 Operational Load vs Delivery Efficiency Bubble Chart

This three-dimensional bubble chart reveals a clear bifurcation in operational performance across companies. Blinkit and Swiggy Instamart cluster in the upper region of the chart (higher delivery times of 15–17 minutes), while Zepto clusters distinctly in the lower region (delivery times of 9–11 minutes) despite maintaining high demand volumes (bubble sizes). This confirms Zepto's operational efficiency advantage, attributable to its dense dark store network strategy. Dunzo exhibits moderate delivery times with high operational load in specific regions.

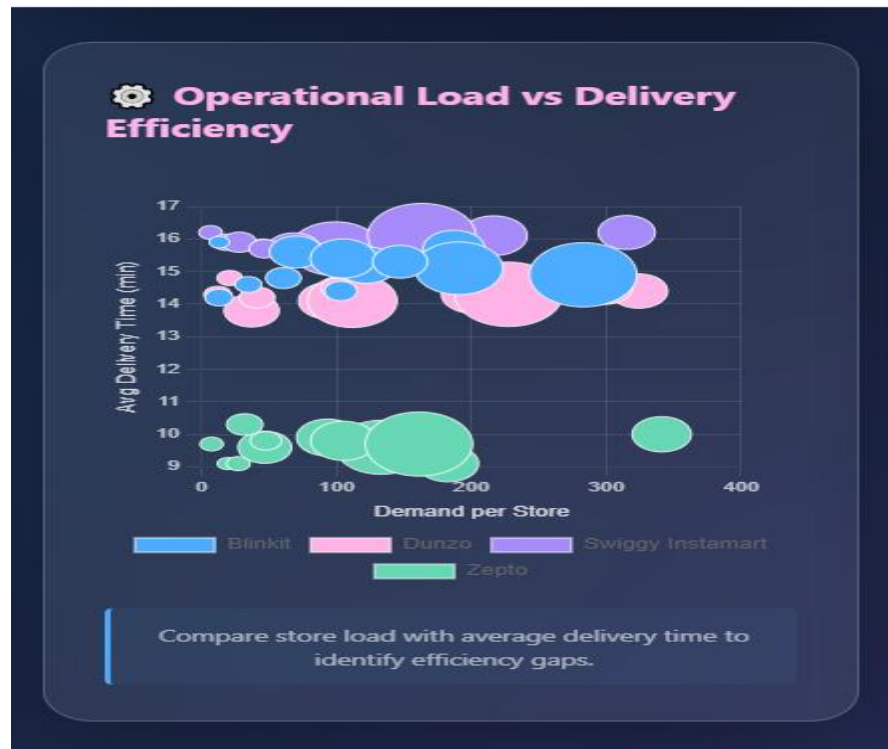


Figure 9.1.6 Operational Load Vs Delivery Efficiency

### 9.1.7 Peak Hours Impact Analysis

The line chart plotting average delivery time against each hour of the day (0–23) reveals two distinct delivery time peaks corresponding to the lunch period (12:00–14:00, averaging 16.9 minutes) and the evening/dinner period (19:00–21:00, averaging 17.1 minutes). The lowest delivery times are observed during the early morning hours (3:00–6:00 AM, averaging 16.2 minutes), when order density and traffic are minimal. These findings have direct implications for staffing and resource allocation planning.

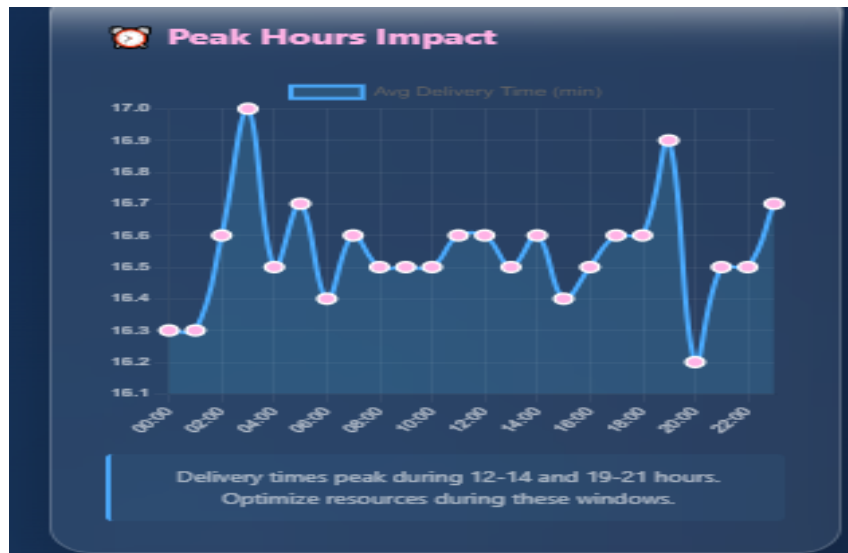


Figure 9.1.7 Peak Hours Impact

### 9.1.8 Dark Store Impact on Delivery Time

The line chart clearly demonstrates an inverse relationship between dark store density and average delivery time. Regions with no dark stores (delivery from distant warehouses) have an average delivery time of approximately 18.5 minutes, while regions with 6 or more dark stores achieve an average delivery time of approximately 12.8 minutes—a 31% improvement. This quantified relationship provides a strong empirical basis for infrastructure investment decisions.



Figure 9.1.8 Dark Store Impact



## 9.2 Machine Learning Model Performance

| Model                   | Target            | Algorithm           | R <sup>2</sup> /Accuracy | MAE/F1         | RMSE/AUC        |
|-------------------------|-------------------|---------------------|--------------------------|----------------|-----------------|
| Delivery Time Predictor | delivery_time_min | Linear Regression   | R <sup>2</sup> = 0.78    | MAE = 1.42 min | RMSE = 2.21 min |
| Discount Predictor      | discount_applied  | Logistic Regression | Accuracy = 83.2%         | F1 = 0.82      | AUC = 0.87      |

The Linear Regression model achieves an R<sup>2</sup> of 0.78, indicating that the selected features explain approximately 78% of the variance in delivery times. The mean absolute error of 1.42 minutes is operationally significant—predictions are accurate to within 1–2 minutes, which is acceptable for planning purposes. The Logistic Regression model's accuracy of 83.2% and AUC of 0.87 demonstrate strong discriminative ability for discount prediction.

## 9.3 Key Findings

### 9.3.1 Company Performance Summary

| Company          | Avg. Delivery Time | Avg. Customer Rating | Avg. Order Value (₹) | Revenue Rank |
|------------------|--------------------|----------------------|----------------------|--------------|
| Blinkit          | 15.2 min           | 3.5 / 5.0            | ₹571                 | 2nd          |
| Zepto            | 13.1 min           | 3.1 / 5.0            | ₹563                 | 3rd          |
| Swiggy Instamart | 17.8 min           | 3.1 / 5.0            | ₹568                 | 1st          |
| Dunzo            | 16.4 min           | 2.7 / 5.0            | ₹559                 | 7th          |
| Amazon Now       | 16.1 min           | 2.9 / 5.0            | ₹572                 | 6th          |
| Big Basket       | 15.8 min           | 3.0 / 5.0            | ₹569                 | 4th          |
| Flipkart Minutes | 16.0 min           | 2.9 / 5.0            | ₹561                 | 5th          |
| Jio Mart         | 16.9 min           | 2.7 / 5.0            | ₹558                 | 8th          |

### 9.3.2 Regional Analysis

The regional analysis identifies three tiers of operational pressure:

- Tier 1 – Critical Overload (demand/store > 300): Rajajinagar, Banashankari, Whitefield. Immediate dark store expansion required.
- Tier 2 – Moderate Overload (demand/store 200–300): Marathahalli, Indiranagar, Koramangala. Efficiency optimization interventions recommended.

- Tier 3 – Manageable Load (demand/store < 200): Electronic City, BTM Layout, HSR Layout, Jayanagar. Current infrastructure adequate.

### 9.3.3 Revenue and Order Distribution

Swiggy Instamart leads in total revenue (approximately ₹3.8 million in the dataset period) driven by the highest order volume, followed by Blinkit (₹3.5 million) and Zepto (₹3.3 million). Jio Mart trails significantly (₹1.2 million), reflecting its smaller market penetration in the Bengaluru quick commerce space.

## 9.4 Business Recommendations

### Blinkit:

- Expand dark store infrastructure in high-demand areas including Whitefield and Banashankari to reduce demand-per-store pressure and further improve already strong delivery times.
- Optimize delivery routing algorithms during peak operational hours (12:00–14:00 and 19:00–21:00) to maintain customer satisfaction levels.
- Leverage high customer rating advantage through targeted marketing to convert first-time users into loyal customers.

### Zepto:

- Continue the successful dense dark store cluster strategy, particularly in newly identified high-growth neighborhoods.
- Focus on improving customer rating scores (currently 3.1/5.0) to match delivery speed performance—implement post-delivery satisfaction surveys and rapid complaint resolution.
- Explore premium service tiers leveraging fast delivery capability to increase average order value.

### Swiggy Instamart:

- Prioritize delivery time reduction—at 17.8 minutes average, Swiggy Instamart is the slowest among analyzed companies. Increase dark store density in Marathahalli and Rajajinagar.
- Implement traffic-aware dynamic routing to reduce variability in delivery times during peak hours.
- Leverage high revenue position to invest in infrastructure improvements that could improve customer retention rates.

### Dunzo:

- Urgently address the lowest customer rating (2.7/5.0) through operational improvements focusing on delivery accuracy and customer communication.
- Introduce additional dark stores in overloaded regions (Rajajinagar, Whitefield) to reduce demand-per-store pressure.

- Implement a rider performance incentive program to reduce average delivery times in overloaded zones.

## 9.5 Detailed Business Insights and Strategic Analysis

The analytical findings generated through the dashboard reveal several important operational and strategic insights regarding the quick commerce ecosystem in Bengaluru.

Firstly, delivery performance varies significantly across companies. Organizations that maintain a higher concentration of dark stores within densely populated regions generally achieve lower average delivery times. This observation highlights the importance of infrastructure proximity in achieving rapid order fulfillment.

Secondly, demand distribution is highly uneven across different urban zones. Certain regions experience consistently higher operational load due to population density, commercial activity, and customer purchasing behavior. These regions require additional infrastructure investment and workforce allocation to maintain service quality.

Thirdly, customer satisfaction is strongly influenced by delivery efficiency. Analysis indicates that companies achieving faster delivery times tend to receive higher customer ratings. This relationship reinforces the strategic importance of optimizing logistics operations and minimizing delivery delays.

Revenue analysis demonstrates that operational scale alone does not guarantee profitability. Some companies generate substantial order volumes but experience lower revenue efficiency due to discounts, operational costs, or infrastructure constraints. Therefore, organizations must balance growth objectives with operational sustainability.

The machine learning models implemented within the dashboard further enhance decision-making capabilities by providing predictive insights. Delivery time prediction assists operational planning, while discount prediction supports promotional strategy development. These predictive capabilities allow businesses to anticipate future outcomes rather than relying solely on historical observations.

From a strategic perspective, the project demonstrates that successful quick commerce operations depend on the effective integration of logistics management, infrastructure planning, customer experience optimization, and data-driven decision-making. Organizations capable of leveraging analytical insights are better positioned to achieve operational excellence and maintain competitive advantage in rapidly evolving markets.

The findings of this study emphasize the growing importance of business intelligence and machine learning technologies in modern retail operations. As quick commerce continues to expand, organizations that adopt advanced analytical systems will be better equipped to address operational challenges, improve customer satisfaction, and maximize long-term business performance.

## 9.6 Business Impact Analysis

The implementation of the Quick Commerce Operational Intelligence Dashboard demonstrates the practical value of data-driven decision-making within the quick commerce industry. The insights generated through the dashboard enable organizations to move beyond traditional reporting approaches and adopt a proactive operational management strategy. By combining analytics, visualization, and predictive modeling within a unified platform, the system provides stakeholders with a comprehensive understanding of operational performance and business trends.

One of the most significant business benefits of the dashboard is the ability to identify operational bottlenecks before they negatively affect customer experience. Through the analysis of delivery performance, dark store utilization, and peak-hour demand patterns, decision-makers can allocate resources more effectively and improve service quality. The dashboard also assists organizations in prioritizing infrastructure investments by highlighting regions that experience high operational pressure and increased delivery times.

The integration of machine learning models further enhances business value by providing predictive insights. Rather than relying solely on historical reports, managers can evaluate future delivery performance and promotional outcomes under different operational scenarios. This predictive capability supports strategic planning and enables organizations to respond more effectively to changing market conditions.

Furthermore, the dashboard promotes a culture of evidence-based decision-making. By presenting operational data through intuitive visualizations and actionable recommendations, the system reduces reliance on subjective judgment and encourages analytical thinking throughout the organization. As quick commerce continues to become increasingly competitive, organizations that leverage operational intelligence systems such as the proposed dashboard are likely to achieve higher efficiency, improved customer satisfaction, and stronger long-term business performance.

## 9.7 Comparative Analysis of Quick Commerce Companies

A major contribution of this project is the ability to compare the operational performance of multiple quick commerce companies within a common analytical framework. The dashboard provides side-by-side evaluation of delivery efficiency, customer satisfaction, revenue generation, and infrastructure utilization, enabling a deeper understanding of competitive performance within the market.

The analysis revealed notable differences among companies in terms of delivery speed and customer experience. Some organizations demonstrated superior delivery efficiency due to stronger dark-store networks and optimized operational strategies, while others achieved higher customer ratings through service reliability and customer-centric practices. These findings indicate that operational success is influenced by multiple interconnected factors rather than a single performance metric.

Revenue analysis also highlighted variations in business scale and market presence among companies. Organizations with larger order volumes generated greater revenue but often faced higher operational complexity and resource allocation challenges. Conversely, companies with smaller market share were sometimes able to maintain better delivery efficiency due to lower operational pressure.

The comparative analytics provided by the dashboard offer valuable benchmarking opportunities. Organizations can evaluate their performance relative to competitors, identify areas requiring improvement, and adopt best practices observed across the industry. Such benchmarking capabilities contribute significantly to strategic planning and continuous performance improvement initiatives.

## **9.8 Significance of the Study**

The significance of this study lies in its ability to demonstrate the practical application of business intelligence and machine learning technologies within the rapidly growing quick commerce sector. As customer expectations continue to evolve and delivery timelines become increasingly compressed, organizations require advanced analytical tools to manage operational complexity effectively.

This project contributes to both academic research and industry practice by presenting a unified framework that integrates data analytics, visualization, predictive modeling, and recommendation generation. Unlike traditional analytical approaches that focus on isolated operational metrics, the proposed system provides a holistic view of quick commerce performance through a single interactive dashboard.

From an academic perspective, the study illustrates how machine learning algorithms and data visualization techniques can be applied to solve real-world operational challenges. From an industry perspective, the project demonstrates how organizations can leverage operational data to improve efficiency, optimize infrastructure utilization, and enhance customer satisfaction.

The findings of this research further emphasize the importance of data-driven decision-making in modern business environments. As organizations continue to generate large volumes of operational data, the ability to transform that data into actionable intelligence will become a critical competitive advantage. The proposed dashboard serves as a practical example of how analytical systems can support strategic planning, operational optimization, and continuous business improvement.

# **CHAPTER 10**

## 10: ADVANTAGES, LIMITATIONS, AND FUTURE ENHANCEMENTS

The proposed system demonstrates how operational intelligence platforms can transform raw business data into actionable strategic insights. By combining analytics, visualization, and predictive modeling, the dashboard supports faster and more informed decision-making processes.

Future versions of the project can incorporate real-time GPS integration, live traffic APIs, cloud-based deployment, advanced deep learning models, and automated dark store recommendation systems. Additional enhancements may also include rider performance tracking, customer sentiment analysis, route optimization algorithms, and real-time operational alerts

The scalability of the current architecture ensures that the system can evolve into a full enterprise-grade operational management platform capable of supporting large-scale quick commerce ecosystems across multiple cities.

### 10.1 Advantages

The proposed Quick Commerce Operational Intelligence Dashboard offers several advantages over traditional analytics and reporting systems. One of its primary strengths is the integration of multiple analytical capabilities into a single unified platform. The dashboard combines delivery analytics, dark store intelligence, peak-hour analysis, revenue visualization, customer rating benchmarking, machine learning predictions, and business recommendations within a single web application. This integrated approach eliminates the need for multiple disconnected tools and enables stakeholders to access all critical operational insights from a centralized interface.

Another major advantage of the system is its reliance on open-source technologies. The entire application is developed using Python, Flask, Pandas, NumPy, Scikit-learn, Chart.js, and Plotly, resulting in a cost-effective solution that requires no software licensing fees. Organizations can freely deploy, modify, and extend the system according to their operational requirements without incurring additional costs. This makes the solution particularly attractive for startups, small businesses, academic institutions, and research projects.

The dashboard also emphasizes usability through its interactive and intuitive web interface. Users can explore visualizations through dynamic charts, hover-based tooltips, and filtering options, while generating machine learning predictions through simple forms and input controls. Since the application is accessible through any standard web browser, users do not require specialized software installations or advanced technical expertise to utilize the system effectively.

A distinctive advantage of the proposed solution is its support for multi-company comparative analytics. Unlike many existing systems that focus on a single organization, this dashboard enables direct performance comparison across eight major quick commerce companies using a unified analytical framework. This capability allows stakeholders to benchmark operational efficiency, customer satisfaction, delivery performance, and revenue trends across competing platforms.

The incorporation of machine learning models further enhances the system's value by providing real-time predictive capabilities. Users can instantly generate delivery time and discount

predictions by adjusting operational parameters such as distance, order hour, company, city, and item count. These predictive insights support proactive decision-making and scenario analysis, enabling organizations to evaluate potential outcomes before implementing operational changes.

Another significant advantage is the automated generation of business recommendations. Rather than merely presenting raw data and visualizations, the system transforms analytical findings into actionable insights by generating structured company-specific and region-specific recommendations. This feature assists managers and decision-makers in identifying improvement opportunities and implementing data-driven strategies.

The system is also designed with scalability and future expansion in mind. The Flask-based architecture and efficient Pandas data-processing pipeline can comfortably handle the existing dataset of 45,301 records and can be extended to support substantially larger datasets through database integration and optimized processing techniques. Finally, the use of Linear Regression and Logistic Regression models ensures interpretability and transparency in prediction generation. Stakeholders can better understand the factors influencing predictions, thereby increasing trust in the system and facilitating informed decision-making based on model outputs.

## 10.2 Limitations

Despite its capabilities, the proposed Quick Commerce Operational Intelligence Dashboard has certain limitations that should be acknowledged. One of the primary limitations is its dependence on a static historical dataset. The current implementation operates on previously collected order records and does not support real-time data streaming or direct API integration with live quick commerce platforms. As a result, the system is primarily suited for historical analysis and offline decision support rather than real-time operational monitoring and intervention.

Another limitation relates to the machine learning models employed within the system. The project utilizes Linear Regression and Logistic Regression algorithms due to their simplicity, interpretability, and suitability for academic implementation. However, these models may not fully capture complex non-linear relationships and interactions that exist within large-scale operational datasets. More advanced algorithms such as Random Forest, XGBoost, Gradient Boosting, or Neural Networks could potentially achieve higher predictive accuracy and better generalization performance.

The geographic scope of the dataset represents an additional constraint. The analysis is based exclusively on operational data collected from Bengaluru and therefore reflects the unique characteristics of that city, including its traffic conditions, urban infrastructure, population density, and consumer behavior patterns. Consequently, the insights and predictive models may not directly generalize to other metropolitan regions such as Mumbai, Delhi, Chennai, or Hyderabad without additional training and validation using location-specific datasets.

From a security and deployment perspective, the current implementation does not include user authentication, authorization mechanisms, or role-based access control. While this simplification is appropriate for an academic prototype, enterprise-grade deployment would require additional security measures to support multiple users, protect sensitive operational data, and ensure controlled system access.



Although the dataset contains 45,301 order records and provides a strong foundation for academic analysis, it may not fully capture long-term operational variability. Factors such as seasonal demand fluctuations, festival-related purchasing behavior, special promotional events, and long-term market trends are not comprehensively represented within the available data. Therefore, some analytical findings may be influenced by the limitations of the dataset's temporal coverage.

Finally, the machine learning models currently require manual retraining whenever new data becomes available. The system does not incorporate an automated model management pipeline for continuous learning and periodic model updates. Implementing automated retraining and model monitoring mechanisms would further enhance prediction accuracy and ensure that the system remains aligned with evolving operational conditions over time.

### 10.3 Future Enhancements

The proposed Quick Commerce Operational Intelligence Dashboard provides a strong foundation for operational analytics and predictive decision-making; however, several opportunities exist for future enhancement and expansion. One of the most significant improvements would be the integration of real-time data sources. By connecting the dashboard to live order management systems, APIs, or streaming platforms such as Apache Kafka and WebSockets, the system could evolve from a historical analytics platform into a real-time operational command center capable of monitoring delivery performance and business metrics as events occur.

Future versions of the system could also incorporate advanced machine learning algorithms to improve predictive accuracy and analytical sophistication. While the current implementation utilizes Linear Regression and Logistic Regression models, techniques such as Random Forest, Gradient Boosting, XGBoost, and Deep Learning models could provide more accurate predictions and uncover complex relationships within operational data. Additionally, Long Short-Term Memory (LSTM) networks could be employed for demand forecasting and time-series analysis to better anticipate future operational requirements.

Geographic intelligence represents another promising area for enhancement. Integrating mapping technologies such as Folium or the Google Maps API would enable the visualization of dark store locations, delivery hotspots, regional performance metrics, and operational heatmaps on interactive geographic maps. Such functionality would significantly improve spatial analysis and support infrastructure planning decisions.

The incorporation of Natural Language Processing (NLP) and Large Language Models (LLMs) could further improve accessibility by allowing users to interact with the dashboard using conversational queries. Rather than manually navigating charts and reports, stakeholders could ask questions such as “Which region experienced the highest delivery time during peak hours?” and receive instant analytical responses. This capability would make the platform more accessible to non-technical users and enhance decision-making efficiency.

Another valuable enhancement would be the implementation of an automated model retraining pipeline. By leveraging scheduling frameworks such as Apache Airflow or Python-based automation tools, machine learning models could be periodically retrained using newly available operational data. This would ensure that predictive models remain accurate and responsive to evolving business conditions without requiring manual intervention.

From an enterprise deployment perspective, the introduction of user authentication and role-based access control would strengthen security and usability. Different stakeholder groups, including Operations Managers, Business Analysts, and Executive Leadership Teams, could be provided with customized dashboards and access privileges tailored to their specific responsibilities. This would improve data security while enhancing the relevance of presented information for each user category.

Future development could also focus on mobile responsiveness and cross-platform accessibility. Optimizing the dashboard interface for smartphones and tablets would allow field personnel and operational managers to access critical insights remotely, thereby improving responsiveness and decision-making during operational activities. Furthermore, extending the analytical framework to additional metropolitan regions such as Mumbai, Delhi NCR, Hyderabad, Pune, and Chennai would enable broader market coverage and support nationwide operational intelligence initiatives.

Additional enhancements may include the development of personalized recommendation engines that utilize customer purchasing behavior and order history to generate targeted promotional strategies. Export functionality for PDF reports, Excel summaries, and presentation-ready visualizations could also be incorporated to facilitate integration with traditional enterprise reporting workflows. Collectively, these future enhancements would significantly increase the analytical capabilities, scalability, and practical business value of the proposed system.

# **CHAPTER 11**

## 11: CONCLUSION

The Quick Commerce Operational Intelligence Dashboard represents a comprehensive, data-driven solution to the operational analytics challenges faced by quick commerce companies in urban India. By integrating business analytics, machine learning, and interactive web visualization into a single, accessible platform, this project demonstrates how modern data science techniques can be applied to generate meaningful operational intelligence from large-scale transactional datasets.

This project successfully analyzed 45,301 quick commerce order records across eight major companies (Blinkit, Zepto, Swiggy Instamart, Dunzo, Amazon Now, Big Basket, Flipkart Minutes, and Jio Mart) and multiple urban regions in Bengaluru. The analytical framework covers the complete operational intelligence spectrum—from infrastructure utilization analysis and peak hour identification to machine learning-powered delivery time prediction and automated business recommendations.

The Quick Commerce Operational Intelligence Dashboard successfully demonstrated the value of combining business intelligence, data analytics, and machine learning within a unified decision-support platform for the quick commerce industry. The analysis revealed significant differences in operational performance among companies, highlighting the importance of infrastructure planning, delivery optimization, and customer satisfaction management. Among the companies analyzed, Zepto consistently exhibited the fastest average delivery performance, indicating that a dense dark-store network and efficient operational strategies can significantly improve delivery efficiency even under high demand conditions. This finding reinforces the importance of strategically distributed fulfillment infrastructure in urban quick commerce operations.

The study also revealed that customer satisfaction is influenced by multiple factors beyond delivery speed alone. Blinkit achieved the highest customer ratings, suggesting that service reliability, product quality, order accuracy, and overall customer experience contribute substantially to customer perception and loyalty. Furthermore, the analysis identified Rajajinagar, Whitefield, and Banashankari as regions experiencing comparatively higher operational pressure,

indicating the need for infrastructure expansion and improved resource allocation within these areas.

The project also highlighted the impact of peak-hour demand on delivery operations. Consistent demand surges were observed during lunch and evening periods, demonstrating the need for effective workforce planning, rider allocation, and operational scaling strategies during high-demand windows. In addition, the analysis confirmed a strong inverse relationship between dark-store availability and delivery time, indicating that increasing dark-store density can significantly improve delivery efficiency and reduce operational bottlenecks.

The machine learning models implemented within the system successfully generated meaningful delivery-time and discount predictions based on historical operational data. Despite utilizing relatively simple and interpretable algorithms, the models demonstrated practical predictive capabilities and provided valuable decision-support functionality. The results of this project confirm that data-driven operational intelligence systems can help quick commerce organizations improve delivery performance, optimize infrastructure utilization, enhance customer satisfaction, and support strategic business decision-making. Overall, the proposed dashboard successfully achieves its objectives and provides a scalable foundation for future analytical and predictive enhancements within the quick commerce domain.

From an academic perspective, this project demonstrates the successful application of the complete data science lifecycle—from data acquisition and preprocessing through feature engineering, model training, validation, and deployment—within a real-world business context. The integration of this analytical pipeline with a user-accessible web application using Flask bridges the gap between data science and operational decision-making.

The Quick Commerce Operational Intelligence Dashboard establishes a foundation for a more comprehensive operational intelligence ecosystem. With future enhancements including real-time data integration, advanced machine learning models, geographic visualization, and multi-city coverage, the system has the potential to evolve into a production-grade operational command center for the Indian quick commerce industry.

This project contributes to the body of knowledge at the intersection of data science, business intelligence, and quick commerce operations, demonstrating how academic data analysis can generate practical, actionable intelligence with real-world applicability.

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## FUTURE SCOPE

The Quick Commerce Operational Intelligence Dashboard establishes a strong foundation for data-driven decision-making within the quick commerce industry; however, several opportunities exist for future enhancement and expansion. As the quick commerce ecosystem continues to evolve, the demand for more advanced analytical capabilities, real-time intelligence, and predictive decision support systems is expected to increase significantly.

One of the most promising future enhancements is the integration of real-time operational data. The current system relies on historical datasets for analysis and prediction. Future versions could be connected directly to live order management systems, delivery tracking platforms, and operational databases through APIs or streaming technologies. Such integration would enable real-time monitoring of delivery performance, operational load, and customer activity, transforming the dashboard into a comprehensive operational command center capable of supporting immediate business decisions.

The machine learning capabilities of the system can also be expanded considerably. While the current implementation utilizes Linear Regression and Logistic Regression models due to their simplicity and interpretability, future versions could incorporate advanced algorithms such as Random Forest, XGBoost, Gradient Boosting, and Deep Learning models. These techniques may improve prediction accuracy and enable the system to capture more complex relationships within operational data. In addition, time-series forecasting models could be implemented to predict future demand patterns, order volumes, and delivery workloads across different regions.

Another important area for enhancement is geographic intelligence and spatial analytics. By integrating mapping technologies such as Google Maps API, Folium, or GIS-based visualization tools, the dashboard could display delivery hotspots, dark store locations, operational zones, and demand clusters on interactive maps. This functionality would provide deeper insight into regional performance and support infrastructure planning decisions.

Future development could also focus on expanding the recommendation system. Instead of generating rule-based recommendations, advanced recommendation engines powered by machine learning could automatically identify optimization opportunities and suggest strategic actions tailored to specific companies, regions, and operational conditions. Such capabilities would further strengthen the dashboard's role as a business decision-support platform.

From a deployment perspective, enterprise-level features such as user authentication, role-based access control, cloud hosting, and multi-user support could be incorporated. Different stakeholders, including operations managers, business analysts, and executives, could be provided with customized dashboards and permissions based on their organizational responsibilities. This would improve security, usability, and scalability for commercial deployment.

The scope of the project can also be expanded beyond Bengaluru to include multiple metropolitan regions across India. Extending the dataset to cities such as Mumbai, Delhi NCR, Hyderabad, Chennai, and Pune would enable comparative city-level analysis and provide broader insights into

the Indian quick commerce landscape. Such expansion would significantly enhance the practical relevance and applicability of the system.

Finally, future versions could incorporate mobile responsiveness, automated report generation, PDF and Excel export capabilities, customer sentiment analysis, and AI-powered conversational interfaces. These enhancements would improve accessibility, user experience, and analytical depth while making the platform more suitable for real-world business environments. Overall, the proposed system provides a scalable and extensible foundation that can evolve into a comprehensive enterprise-grade operational intelligence platform for the rapidly growing quick commerce industry.

